

PRISON ENTREPRENEURSHIP TRAINING AND RECIDIVISM

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Abstract

Formerly incarcerated individuals face significant economic and social challenges upon release, with more than half returning to prison within three years. We integrate studies on entrepreneurship training with those that examine the potential of entrepreneurship to overcome labor market discrimination, and investigate whether entrepreneurship training can help reduce recidivism. First, using proprietary data from the Texas PEP program, we compare recidivism outcomes for individuals who attended the program with those who were selected to attend, but could not, due to parole release. Second, we utilize survey data from program graduates to examine the relationship between entrepreneurial entry and recidivism, conditional on having completed the program. Third, we discuss our analysis of qualitative interviews with program graduates and directors. We find significantly lower recidivism rates for individuals who attended the program, and among attendees, significantly lower recidivism rates for those who entered entrepreneurship. Interviews suggest three roles of entrepreneurship in training: vision, focused development, and control. Our findings point to the potential role of entrepreneurship training in settings that involve disadvantaged populations such as justice-impacted individuals.

INTRODUCTION

Entrepreneurship is often considered a path to achieve social mobility (e.g., Sorensen and Sharkey 2014, Rider et al. 2023). This opportunity may be especially salient for disadvantaged entrepreneurs, who may encounter structural barriers in traditional labor markets and broader society (e.g., Rindova et al. 2009, Thebaud 2015, Hwang and Phillips 2024, Yang et al. 2024). At the same time, aspiring entrepreneurs from disadvantaged backgrounds also tend to lack access the critical resources to realize entrepreneurial opportunities, such as financial capital and know-how (Chatterji and Seamans 2012, Kanze et al. 2018, Clough et al. 2019). For this reason, scholars have been inquiring whether provision of entrepreneurial resources to disadvantaged communities through accelerators and entrepreneurship training programs with geographic or demographic targets could address barriers to entrepreneurial entry and success (McKenzie and Woodruff 2014, Dutt et al. 2016, Lyons and Zhang 2018, Assenova 2020).

In this study, we integrate these literature streams to examine the effectiveness of entrepreneurship training programs in a novel setting: formerly incarcerated individuals. In 2022, approximately 1.9 million Americans were incarcerated in state or federal prisons or local jails, with almost 500,000 individuals released from state and federal prisons each year (Carson 2023). Incarcerated individuals face severe difficulties in reintegrating into society upon release, with over 60% returning to prison within three years (Alper et al. 2018). The cycle of recidivism prevents individuals from contributing to society and the estimated cost of incarceration to the US is \$80 billion a year (Smith 2016). A key reason high rates of recidivism is that discrimination against formerly incarcerated individuals in the labor market leaves them with fewer job opportunities and with lower wages when employed (Waldfoegel 1994, Western and Beckett 1999, Western 2002, Pager 2003, Zhang et al. 2021, Agan et al. 2024). Given these employment barriers, entrepreneurship may serve as an alternate path toward economic mobility and socioeconomic standing. A recent study suggests that formerly incarcerated individuals tend to pursue entrepreneurship at higher rates than similar non-incarcerated individuals and they are more likely to become entrepreneurs in states with higher structural barriers to gainful employment (Hwang and Phillips 2024). At the same time, many formerly incarcerated

individuals do not have the requisite training or resources to succeed as entrepreneurs. In recent years, several entrepreneurial training programs for incarcerated individuals have emerged, with the objective to support entrepreneurial entry and reintegration (ICIC 2018). While the initial evidence from prison entrepreneurial programs is encouraging, studies often confound selection into programs with the effect of training (Smith 2016). Thus, we do not yet understand the true impact of entrepreneurship training on outcomes of interest for incarcerated individuals such as recidivism or entrepreneurial activity.

The usefulness of entrepreneurship training programs in supporting disadvantaged entrepreneurs is not a foregone conclusion for two reasons. First, prior work reports mixed findings on the impact of entrepreneurship training programs on entrepreneurial entry and performance (McKenzie and Woodruff 2014, Camuffo et al. 2020, Eesley and Lee 2021). Recent work suggests that disadvantaged entrepreneurs with limited prior experience in entrepreneurship may especially benefit from training (Dutt et al. 2016, Lyons and Zhang 2018, Assenova 2020, Eesley and Lee 2021). However, others report the limitations of similar specialized efforts when they do not consider the unique needs of the populations they serve or cannot address broader patterns of discrimination in entrepreneurial resource provision (Quinn and Woodruff 2019, Lall et al. 2020, Kim and Kim 2022, Bruton et al. 2024, Pongeluppe 2024, Sunao and Rocha 2025). Second, even if such programs can facilitate entrepreneurial entry, disadvantaged entrepreneurs may continue to experience structural barriers and stigma, and may not successfully reintegrate into the society (Pongeluppe 2024, Sunao and Rocha 2025). Coupled with findings on the high rates of failure in entrepreneurship (Knott and Posen 2005, Astebro et al. 2011, Yang and Aldrich 2012), and heterogeneity in entrepreneurial entry (Glaeser et al. 2015, Bennett and Chatterji 2023), it is not clear that entrepreneurship can sufficiently support those who face barriers for gainful employment.

In this paper, we investigate whether entrepreneurship training is associated with decreased recidivism among formerly incarcerated individuals, and if so, why. In doing so, we adopt a broader definition of entrepreneurship that includes self-employment (Hwang and Phillips 2024, Sunao and Rocha 2025). To answer this question, we obtained a unique data set that contains all incarcerated individuals from 2009 to 2016 who were accepted into the Texas Prison Entrepreneurship Training Program (PEP), a

leading prison entrepreneurial training program in the US. Our data includes recidivism outcomes for program participants as well as those who were admitted to the program but did not receive training due to parole release, enabling a direct comparison of trained individuals against a counterfactual of similar others. Our analyses show that formerly incarcerated individuals who received entrepreneurial training experience lower recidivism than their counterparts who did not receive training, and a key mechanism for the difference is skill acquisition rather than network resource provision or certification. While we are not able to observe entrepreneurship outcomes for those released on parole, we utilize post-release survey data of prior program attendees to investigate, conditional upon being trained in the program, whether those who have started businesses experienced lower recidivism rates compared to those who did not. Finally, we examine the pathways through which entrepreneurship training could reduce recidivism with interviews with former program participants.

This study aims to contribute to the literature that expands research on entrepreneurial training programs to include programs for disadvantaged entrepreneurs and marginalized populations (Banerjee et al. 2015, Dutt et al. 2016, Assenova 2020, McKenzie 2021). Learning about programs that target incarcerated individuals can improve our understanding of the richness and varied potential impact of entrepreneurial training programs. In particular, our research expands prior work on training programs by evaluating the impact of entrepreneurial training on recidivism, a new and socially important outcome outside of entrepreneurial activity. By doing so, we highlight the expansive role that entrepreneurship training can play outside of profit maximization. Additionally, we complement a recently emerging literature on formerly incarcerated individuals in entrepreneurship and organizations (Toubiana 2020, Hwang and Phillips 2024) by examining the ways in which formerly incarcerated individuals can be effectively reintegrated into society through training. Our work complements earlier studies that expand entrepreneurship research beyond high-technology, high-growth settings to disadvantaged populations (e.g., Rindova et al. 2009, Thebaud 2015, Hwang and Phillips 2024, Yang et al. 2024, Contigiani et al. 2025). Finally, given the significant challenges facing society due to mass incarceration, understanding whether these types of programs can help reduce recidivism has the potential to shape policy while

increasing the rate of entrepreneurial entry. We aim to provide a novel example of the way that entrepreneurship training can be used as a lever to address grand challenges of our day.

THEORY

In order to understand the relationship between entrepreneurship training and lower recidivism rates, we first review the evidence on whether entrepreneurship training can effectively increase rates of entrepreneurial entry and sustained performance (survival and success) among participants. Second, we review the literature examining the relationship between entrepreneurship and socioeconomic mobility among disadvantaged populations. If indeed we can expect a training program to facilitate successful entrepreneurship and in turn entrepreneurial entry to circumvent structural barriers faced by disadvantaged individuals, we may expect a training program to have a meaningful effect on recidivism rates upon release.

Entrepreneurship Training Programs

An emerging stream of management studies examines the effectiveness of entrepreneurship training in various contexts. Entrepreneurial training could take place in programs that vary in length, content, and structure, including mentoring and advice (Chatterji et al. 2019), training within accelerators and incubators (Dutt et al. 2016, Assenova 2020, Hallen et al. 2020), and short-term entrepreneurship training programs (Camuffo et al. 2020, McKenzie 2021, Kotha et al. 2023, Novelli and Spina 2024). They may be provided by private or social enterprises (e.g., Hallen et al. 2020, Åstebro and Hoos 2021), universities (Eesley and Lee 2021), or sponsored by governments (Armanios et al. 2017, Gonzalez-Uribe and Leatherbee 2018, Leatherbee and Katila 2020). While some are geared toward entrepreneurs in high-technology, high-growth settings (e.g., Hallen et al. 2020), others are targeted toward disadvantaged populations who are more likely to engage in necessity entrepreneurship due to a lack of labor market opportunities or those who are in developing countries and lack access to resources and know-how (e.g., McKenzie and Woodruff 2014, Assenova 2020, McKenzie 2021).

Entrepreneurship programs seek to help nascent entrepreneurs by offering them a combination of knowledge, financial assistance, and certification (Hallen et al. 2020, Hallen et al. 2023). The main objective of many such programs, particularly those offered to disadvantaged populations, is to facilitate entrepreneurial entry and continued survival through teaching effective management practices (McKenzie and Woodruff 2014, McKenzie 2021). Studies report mixed results of programs on entrepreneurial entry, with the highest benefit observed for those without prior entrepreneurial experience (Banerjee et al. 2015, Lyons and Zhang 2018, Eesley and Lee 2021). Some entrepreneurial programs teach aspiring entrepreneurs to critically evaluate business opportunities, which may in fact decrease rates of entrepreneurial entry, while increasing the performance of those businesses started. Indeed, among those who have already started businesses, program attendance is often associated with higher exit, along with positive outcomes in terms of profitability, growth, and access to external capital for those who remain (Chatterji et al. 2019, Camuffo et al. 2020, Hallen et al. 2020, Lall et al. 2020, Kotha et al. 2023).

The findings regarding entrepreneurial entry and performance for trainees from disadvantaged backgrounds and those located in developing regions are mixed. On the one hand, some studies report increased entrepreneurial entry and performance, since participants are often exposed to the knowledge and skills provided by the entrepreneurship training programs for the first time (Banerjee et al. 2015, Assenova 2020, McKenzie 2021, Agarwal et al. 2024). On the other hand, the benefits might be limited (Van der Sluis et al. 2005, McKenzie and Woodruff 2014), especially if the program content does not adequately reflect the structural challenges experienced by entrepreneurs, such as resource access barriers or stigma (Quinn and Woodruff 2019, Lall et al. 2020, Kim and Kim 2022, Bruton et al. 2024, Pongeluppe 2024) or if the skill gap is too large for entrepreneurs to implement the advice and training provided in the program (Otis et al. 2023). Finally, some studies investigate the need to supplement typical business training for disadvantaged entrepreneurs with other skills such as a personal growth mindset, social skills training, and personal initiative training (Patzelt et al. 2014, Campos et al. 2017, Åstebro and Hoos 2021, Dimitriadis and Koning 2022).

Prison Training Programs

The works described above are informative in understanding the effectiveness of entrepreneurship training for disadvantaged populations. At the same time, formerly incarcerated individuals may be different from other disadvantaged populations studied in prior work. First, incarcerated individuals often face a potentially larger knowledge and skills gap than most other trainees studied in the literature. Incarcerated individuals often have lower educational attainment and employment history than the general population, and the gap is exacerbated by skill erosion during sentencing (Davis et al. 2014). Second, the prison environment is a good example of a ‘total institution’ (Goffman 1961), which often imprints individuals with a unique set of norms and ways of thinking. This may constrain individuals’ ability to benefit from entrepreneurship training due to limited trust in program providers, limited motivation to complete the program, or limited beliefs in their ability to succeed as entrepreneurs (Patzelt et al. 2014, Rogers et al. 2017, Toubiana 2020). Third, the discrimination and stigma that affects formerly incarcerated individuals may pose unique challenges in entrepreneurial careers as well, preventing access to resources such as financing or customers (Pager 2003, Western and Pettit 2005, Agan et al. 2024).

Despite these challenges, a few streams of literature within and outside of management research find that prison training programs (not limited to entrepreneurship) may lead to various positive outcomes. The programs studied in prior work include skills training, psychological development training and other forms of in-prison training. A meta-analysis by the RAND corporation found that correctional education reduces the risk of post-release incarceration by 13 percentage points and that every dollar spent on prison education saves the government \$4 to \$5 in re-imprisonment costs (Davis et al. 2014). In addition, prison training programs result in an increase in post-release employment and higher average wages for the trained (Steurer et al. 2001, Davis et al. 2014).

Within the broader literature examining prison training programs, a smaller stream examines the provision of entrepreneurial training to incarcerated individuals (e.g., Cooney 2012, Johnson et al. 2014, Patzelt et al. 2014, Brophy et al. 2024). These studies typically evaluate programs qualitatively or

examine historical data using correlational designs. The findings are mixed, and benefits may depend on participant characteristics, such as optimism (Patzelt et al. 2014). A key challenge in studying the effectiveness of entrepreneurship training programs in the prison context is the availability of data about such programs and individual-level outcomes over time. As a result, studies often rely on small samples and are unable to address selection bias: entrepreneurial training may cause reduced recidivism for individuals with higher innate ability, and thus a lower probability to recidivate (English, 2016). The findings are suggestive, but not conclusive, of the ability for entrepreneurial training to reduce recidivism.

While prior research does not clearly establish the impact of prison training, many of the mechanisms highlighted in prior research on entrepreneurship training are likely to play a role in the relationship between prison entrepreneurship training and recidivism. Primarily, entrepreneurship training programs can provide knowledge and capabilities to realize an entrepreneurial opportunity, enter entrepreneurship, and run a business (Chatterji et al. 2019, Assenova 2020, Hallen et al. 2020). If these skills remove some of the barriers individuals face upon release and increase the likelihood to start a successful business, they could facilitate the integration of the individual to the society, reducing recidivism (Hwang and Phillips 2024). In addition, such programs could provide access to additional resources necessary for successful entrepreneurship, such as legitimacy, social ties with peers, mentors, program staff, and outsiders (Clough et al. 2019, Lall et al. 2020, Hallen et al. 2023). Given the strong stigma and discrimination faced by formerly incarcerated individuals, such resources may play an important role in reintegration (Visser and Travis 2003). Finally, an entrepreneurship training program could help by serving as a turning point that changes the mindset and identity of attendees such that they can alter their long-term trajectory toward legitimate activities (Patzelt et al. 2014, Rogers et al. 2017, Sampson and Laub 2017, Toubiana 2020, Zhang et al. 2021).

Entrepreneurship and Recidivism

If indeed an entrepreneurial training program can help potential entrepreneurs start and successfully run entrepreneurial ventures, the second question is whether such entrepreneurial activity could lead to higher socioeconomic wealth and integration into the society. Indeed, prior work provides

compelling evidence that entrepreneurship can counteract labor market discrimination and provide an alternative to disadvantaged groups who lack attractive employment options or are otherwise penalized in the labor market (Rindova et al. 2009, Thebaud 2015, Dencker et al. 2021, Hwang and Phillips 2024, Yang et al. 2024).

At the same time, others find that entrepreneurship may not be the ‘silver bullet’ that provides a solution to the structural barriers faced by formerly incarcerated individuals. Among those who exhibit entrepreneurial intent, there is often considerable heterogeneity in successfully completing the steps to start a business (Glaeser et al. 2015; Bennett & Chatterji, 2019). In addition, even for typical aspiring entrepreneurs, the process of entrepreneurial entry is fraught with high uncertainty and high rates of failure (Knott and Posen, 2005; Yang & Aldrich, 2017; Astebro et al. 2014; Clough et al 2019). Finally, entrepreneurship does not always successfully address the layers of discrimination and stigma faced by disadvantaged groups, only presenting them with additional challenges in securing resources and running a business (Fairlie et al. 2022; Pongeluppe, 2024). A recent study of formerly incarcerated individuals in Denmark finds that those who entered entrepreneurs earned less and were likely to return to prison compared to those who were employed (Sunao & Rocha, 2024).

In sum, entrepreneurial training can reduce recidivism of formerly incarcerated individuals if it can effectively provide them with the knowledge and skills to start a business, and if entrepreneurial entry can in turn facilitate reintegration into the society by counteracting labor market discrimination. While there is some evidence supporting both conditions, there are also conflicting findings and open questions about each. As a result, we seek to answer the following research question empirically:

Research Question: Do formerlyincarcerated individuals who received entrepreneurial training experience lower recidivism than those who did not?

The subsequent question is if we observe prison entrepreneurship training to be related to lower rates of recidivism, whether this relationship is due to increased entrepreneurial entry. In other words,

does attending an entrepreneurship training program increase entrepreneurial entry rates among formerly incarcerated individuals, and among those who have received training, are those who start businesses less likely to recidivate than those who enter employment and those who do not join the workforce? The answers to these questions are not clear. On the one hand, an entrepreneurship training program might primarily benefit those who intend to start a business by providing the knowledge and resources that specifically facilitate entrepreneurship. On the other hand, the knowledge, resources, and legitimacy provided through entrepreneurship training may be valuable for reintegration into the society, regardless of whether the individual starts a business or not, since the programs can address stigmatization, lack of social support, economic problems, and lack of employment opportunities in multiple ways (Pezzin 1995, Uggen 2000, Western 2002, Visher and Travis 2003). If so, entrepreneurship training could reduce recidivism through employment as well. We investigate these possibilities through exploratory analyses.

CONTEXT: TEXAS PRISON ENTREPRENEURSHIP PROGRAM

The context for this study is the Texas Prison Entrepreneurship Program (PEP). Based in Texas, this program seeks to reduce recidivism and create economic opportunities through training and re-entry services (ICIC 2018). Founded in 2004, the program is one of the largest, most comprehensive, and most influential training programs of its kind, training and serving hundreds of incarcerated individuals each year in partnership with the Texas Department of Correctional Justice (See Appendix A for comparison with other programs). Below we briefly describe the process of recruitment, selection, and training at PEP. More details about the selection and training of individuals are in Appendix B.

Every eligible incarcerated male in Texas receives an invitation letter from PEP to apply to the program when there are three years left in their sentence¹. In addition to being within three years of release, an individual must have or be in the process of attaining a high school diploma or GED, and not be an active gang member or sexual offender to be eligible. Each quarter, PEP sends approximately 2,000

¹ A comparable program for female incarcerated individuals has recently started.

recruiting letters to eligible incarcerated individuals and receives about 600 applications. All applications are evaluated based on the quality of the responses and perceived likelihood of exerting sufficient effort to complete the program. Applicants are often interviewed by program staff prior to acceptance. Acceptance decisions are made based on a risk scoring system designed to select incarcerated individuals with the greatest likelihood to graduate. Out of the approximately 600 applicants, about 160 are accepted to the program each quarter. Accepted candidates are relocated to one of two facilities designated to work with the program. These participants receive a 3-month life skills training program called “Leadership Academy” and a 6 month “mini-MBA” designed to help them become successful entrepreneurs and productive citizens upon release. Graduates receive a certificate of entrepreneurship from the Hankamer School of Business of Baylor University. The program also enables development of social ties with peers, mentors, program staff and outsiders through numerous social activities such as mentoring sessions, the business plan competition and graduation day. Finally, the program provides a reintegration program consisting of transitional housing, a temporary living stipend, job placement assistance, and a care package that is given to any graduate that chooses to receive these benefits upon release. In this way, the program provides human, social, and financial capital as well as certification to aspiring entrepreneurs (Clough et al. 2019).

METHODS

Study 1: PEP Training and Recidivism

Data

To examine the relationship between entrepreneurial training and recidivism, we have obtained a novel data set that contains information on all incarcerated individuals who applied and were accepted to PEP between 2009 and 2016, including those who are accepted but did not receive training. Out of the 2,741 individuals who were admitted to the program, 969 received training, and 1,783 did not due to health issues or early release on parole, even though they had qualified to join the program in the selection process. We exclude five individuals who were deceased after release and 233 individuals who could not

attend due to health issues, since we do not know the severity of health conditions and the implications thereof. The final sample includes 2,509 individuals, 969 of whom have attended the program and 1,540 who were released on parole prior to receiving training.

A naïve comparison of outcomes for incarcerated individuals who have received training with all those who have never received it is subject to selection concerns, as individuals who have received training may be different from those who did not in terms of innate ability or entrepreneurial drive. In order to alleviate this concern, we compare the recidivism outcomes for individuals who have attended the program with those who were accepted to join the program but did not do so due to receiving parole. Both groups undergo the strict selection process conducted by program staff, as described above. After a candidate has been accepted, they must wait until the Texas Department of Correctional Justice authorizes their physical transfer to the PEP prison facilities, which can take months. During this time, a proportion of incarcerated individuals who have qualified to join the training program receive sentencing reduction on parole and are thus released from prison before the training can take place. We assume that the program selection criteria ensure that the admitted individuals are equally motivated and able to succeed in and after the program, and we can therefore compare recidivism outcomes for individuals who attended the program with those who were admitted but did not attend. We have validated this assumption with multiple interviews with PEP leadership. (See Appendix C for details on the parole process and further justification of the assumption). Potential selection bias is reduced since all individuals in the dataset have already applied, been screened, and accepted to the program, and have demonstrated similar levels of interest in entrepreneurial training, effort to apply, and qualifications.

Dependent variables

The dependent variable is recidivism, or the reincarceration of an individual after release. If an individual was incarcerated multiple times after first release, we use the date of first reincarceration to determine recidivism date. These data were provided by the Texas Department of Correctional Justice. We first use a dummy variable that captures 3-year recidivism, which is the incidence of crime that leads to reincarceration within three years of release. The 3-year time period is standard in criminology studies and

is a primary determinant of lifelong propensity to recidivate over longer periods of time (Steurer et al. 2001, Davis et al. 2014). In addition this variable provides a standard time window to measure recidivism regardless of different release years. We also predict total recidivism, which is the incidence of reincarceration at any time between program attendance and the end of the study period. The data on recidivism outcomes were obtained from the Texas Department of Correctional Justice records and were up to date as of Dec 31, 2020.

Independent variable

The independent variable is an indicator variable showing whether an accepted individual received PEP training or not. We code a dummy variable that identifies all individuals who have received PEP training².

Control variables

Since PEP does not collect comprehensive data on individuals who did not participate in training, we have access to few observable characteristics of untrained individuals in the sample. We were able to control for race with indicator variables for Black, Hispanic and Other (white is the omitted category). We also controlled for age of individuals at acceptance with indicator variables for four age groups representing 10-year bins (under 25 is the omitted category). We control for release year by including year dummies between 2010 and 2016.

Estimation Method

Since the dependent variable (recidivism) is an indicator variable, we used logistic regression models to estimate the relationship between training and recidivism. All models include release year dummies as well as clustered standard errors. Because the dataset is a pooled cross section in which each individual appears once, we cannot use individual fixed effects. We report robustness tests with linear probability models and Cox proportional hazard models in the Appendix.

When we tested sample balance with respect to race and age group, we found imbalances between treated and control samples. We have interviewed program leadership to verify that this

² The data set does not contain information on any individuals who started the program but did not complete it, either due to parole or other reasons, in the time window for which we were able to obtain recidivism data.

imbalance is not systematic. We have also found that the imbalance was not consistent across cohorts. Based on our interviews and examination of the data, we have no reason to believe that parole is unevenly distributed across races or age groups. Still, we have employed coarsened exact matching (CEM) to achieve a higher sample balance, matching individuals in the treated and control samples based on race, age group, and release year. This procedure can address concerns about selection to the extent that these observable characteristics are correlated with unobservable characteristics (Iacus et al. 2012).

RESULTS

While we have access to only a few observables, we can evaluate the similarity between treatment and control groups based on race and age (Appendix D for comparison tables). First, almost half of accepted individuals are in the 25-34 age group and almost 65% of individuals belong to racial minorities, patterns that are consistent with extant research (Western 2002, Pager 2003, Hwang and Phillips 2024). Following the CEM procedure, the sample imbalance is not eliminated, however, it is reduced, as L1 distance declines from 0.346 to 0.265 (Table 1). The number of observations declines from 2,509 to 2,461.

Table 2 presents a cross tabulation of recidivism rates for the trained and untrained groups before matching. The untrained group experienced a 22.14% rate of 3-year recidivism, while the trained group experienced an 8.67% rate of 3-year recidivism. The difference in means is 13.47% ($t = 8.894$, $p = 0.000$). Similarly, there is a significant difference in rates of total recidivism between the untrained (32.79%) and trained (15.99%) groups ($t=9.465$, $p = 0.000$).

Table 3 presents the results of the logistic regression models with both dependent variables (3-year and total recidivism) before matching (Models 1-4) and after CEM (Models 5-8). The results are consistent through all models, with and without the addition of control variables. According to margins calculated based on Models 3 and 4, trained individuals have 0.088 probability of recidivating within three years and 0.161 probability of recidivating through the study period, whereas the corresponding average probabilities for untrained individuals are 0.219 and 0.326. The 3-year and total recidivism rates for the untrained individuals are 2.48 and 2.02 times those for trained individuals, respectively. We also

observe based on predictive margins on Model 3 that white individuals have a 0.20 average probability of recidivating within three years, whereas Black and Hispanic individuals have 0.156 and 0.149 average probability. The results after CEM (Models 5-8) are consistent, with minimal decrease in coefficient estimates. These results suggest that attending PEP has a dramatic impact of subsequent recidivism. We repeated these analyses with linear probability models (LPM), Cox proportional hazard models, and alternate matching methods (Black et al. 2020) and obtained consistent results (Appendix D).

Supplementary Tests

The key assumption of the study is that the individuals who were selected but were unable to attend the program due to parole are not systematically different from those who were able to attend. This assumption could be violated if individuals with certain characteristics were more likely to qualify for parole or they faced different conditions following release, regardless of program attendance. As we explained in detail in Appendix C, this is unlikely since individuals are carefully screened for parole eligibility and the reported rates of recidivism for parolees in Texas was lower than the general prison population (20.2% vs. 25.6%). However, we tested the robustness of our results with another group of applicants who were admitted but were not able to attend the program: Those who were discharged from prison due to health concerns ($n=232$). We find that the mean differences in recidivism rates between parole and health groups are not statistically significant and analyses with medical discharge as the comparison set provide similar results to the reported ones (Appendix D).

Next, we investigated the mechanisms underlying these strong results. Prior literature suggests multiple pathways that could explain the relationship between program attendance and recidivism. First, participants may benefit from gaining access to the program's network of participants and volunteers, which may help them in their post-release endeavors. Second, they may benefit by gaining legitimacy through certification. Earning a certificate through program completion can provide them with legitimacy, help them overcome stigma and reintegrate into the society. Third, they may benefit by applying the lessons they have learned in the program. Finally, they may simply benefit from disassociating with former relationships, as embeddedness in criminal networks may propagate repeat offenses (Hagan 1993,

Visher and Travis 2003). In order to probe into these possibilities, we conducted some exploratory analyses. First, we investigate whether program attendants primarily benefit through access to the program network. While we do not have sufficient data to measure each individual's access and ability to leverage the network, we make the assumption that the program network grows over time as more participants graduate and program success attracts more volunteers. We also assume that the program network benefits individuals the most when they are first released, since this is a vulnerable time in which they may not have resource access. This implies that the more recent graduates of the program should benefit more from network access than earlier graduates, since the network grows over time. We test for this possibility by examining 3-year recidivism rates by graduation year (Figure 2). We find no sustained increase in the difference in recidivism rates between the trained and untrained individuals over time, and in fact, a decline after 2014. This pattern is inconsistent with the explanation that the network benefits account for the results. Second, we examine the legitimacy and certification pathway. If the program primarily benefits participants through conferring legitimacy, we may expect those with the highest need for legitimacy to benefit from the program the most. As prior research suggests that the stigma of incarceration is higher for Black compared to white individuals (Pager 2003), we examine the rate of change in recidivism upon program attendance by race by including an interaction of training with race (Appendix D for LPM results). As visualized in Figure 3, white attendees experienced a larger decrease in recidivism than others, which is inconsistent with a legitimacy-based explanation for program effectiveness. Finally, if the program benefits individuals primarily through teaching them business and life skills, we may expect younger individuals who are likely to have lower experience to benefit more from the program than older individuals. We explore this possibility by including the interaction of training with age group. As visualized in Figure 4, the results indeed suggest that individuals under 25 who attend the program experience the largest drop in recidivism compared to those who did not attend (Appendix D for full results). This pattern is consistent with the possibility that a primary benefit of the program is through learning rather than network relationships or legitimacy. Next, we investigate the role of disassociating from former relationships. PEP provides a reintegration program to graduates that

provides benefits such as temporary housing, medical assistance, and job placement help to graduates of the program (Guillot 2014), with the aim of facilitating reintegration into the society and preventing graduates from ‘falling back into their old ways’. However, not all individuals choose to receive this support. This program does not have an educational component and only provides post-release assistance. Indeed, in our interviews, subjects discussed the importance of the reintegration program in helping them establish their new lives. One subject discussed how he could have avoided returning to his old network and resuming drug abuse had he used the program. We therefore use participation in the reintegration program as a proxy for disassociating from old networks. We repeated the logit analyses reported in Table 3 by splitting those who were trained into two groups: those who received reintegration services and those who did not. The results suggest that reintegration services play a large role in reducing recidivism (Appendix D). According to predictive margins, graduates who complete the reintegration program have a 5.9% (12.9%) probability of 3-year (total) recidivism, graduates who did not complete the reintegration program have a 10.5% (18.1%) probability of 3-year (total) recidivism. Both groups exhibit significantly lower recidivism than the accepted but untrained individuals (21.9% and 32.6% probability of 3-year and total recidivism rates, respectively), suggesting that the training received in the program as well as post-release assistance are important in reducing recidivism. Therefore, our findings are consistent with learning effects as well as the importance of breaking out of old social relationships.

Study 2: Post-Release Status and Recidivism

An important question is whether the entrepreneurship training program reduces recidivism through increased entrepreneurial activity. Unfortunately, Texas PEP and the Texas Department of Correctional Justice were unable to provide us with detailed data beyond race and age on individuals who have never joined the program, limiting our ability to compare entrepreneurship rates in the trained and untrained groups. However, the program does conduct an annual survey of graduates, which allows us to track the post-release activities of survey respondents after release. While we cannot answer the question of whether program attendance increases the rate of entrepreneurship, we can examine what proportion of

survey respondents started businesses and whether survey respondents who started businesses were less likely to recidivate than other respondents.

We were able to obtain data on 505 respondents of the graduate survey who were in our archival data set. This comprises 52.1% of the 969 program attendees in our earlier analysis, higher than the 36% average response rates in organizational studies (Baruch and Holtom 2008). While the survey has been conducted annually since 2013, not all respondents have answered the survey in all years. We have used the most recent survey filled by each respondent to study post-release outcomes. The mean difference between the release date and survey date was four years. We have checked for the robustness of our results to excluding survey responses that were up to three years from release date and found similar results. We compared survey respondents to the program attendees in the archival study and found no statistically significant differences in terms of age ($t = -0.947$), belonging to a racial minority ($t = -0.797$), and release year ($t = -1.482$).

We have identified entrepreneurial activity through the answer to the survey question, “Have you started a business since your release?” Out of the 505 respondents, 185 (36.6%) have reported starting a business and 239 (47.3%) have reported being employed. The entrepreneurial entry rate is significantly higher than the 13.31% entrepreneurial entry rate reported by Hwang and Phillips (2024) for formerly incarcerated individuals based on the 1997 US National Longitudinal Survey of Youth sample as well as the 4-5% average rate of entrepreneurship among U.S. households reported by Salgado in the late 2010s (2020)³. We coded the remaining 16.8% as unemployed.

Of those who started a business, 158 (85%) have reported that their business is still active. The most common business types were construction (16.2%), transportation (12.3%), home repair and maintenance (10.06%), landscaping (8.94%), health, wellness, and beauty (7.82%), and automotive (6.15%). There was a wide variety in the business types reported, including engineering, technology, real estate, and design. Almost 85% of businesses employed at least one person, and only 11 employed more

³ Even if the entrepreneurial entry rate among 464 non-respondents was zero, the rate of entrepreneurship would be $185/969=19.1\%$, which is still higher than the reported rates in prior work.

than five. About 85% continued to operate their businesses at the time of the survey (Detailed summary statistics provided in Appendix D).

Table 4 shows the cross-tabulation of entrepreneurship and employment with recidivism in the survey subsample. Accordingly, 3-year recidivism rates are 2.22% for entrepreneurs, 5.44% for those who were employed, and 3.53% for those who were unemployed. The differences between groups are not statistically significant. Total recidivism rates are 4.3% for entrepreneurs, 8.4% for those who were employed, and 17.6% for those who were not employed. The mean rates for both entrepreneurs and those who were employed are statistically different from the mean rates for the unemployed. It is interesting that the mean recidivism rate for entrepreneurs is about half of the rate for the employed.

We further investigate the relationship between entrepreneurship, employment, and probability to recidivate through LPM with robust standard errors in Table 5. We include the earlier controls for race, age, and release year. Models 1 and 2 show that compared to those who were unemployed, entrepreneurs have a significantly lower 3-year and total recidivism rates. The relationship between employment and 3-year recidivism rates are not significant in Models 3 and 4. Compared to the unemployed, the employed individuals have lower rates of total recidivism in the full model and the differences are significant (Model 6).

In sum, we find that those who started businesses after completing the program were less likely to recidivate than others, and those who were employed had significantly lower total recidivism rates than those who were not employed. However, the analysis potentially has two important limitations. First, we do not have entrepreneurship and employment data for the 47% of attendees who did not fill the survey, which could cause sample selection bias. The second issue is that those who choose entrepreneurship may be different in unobservable ways from those who did not choose entrepreneurship as a career path, causing an omitted variable bias. We report the results of additional analyses addressing these limitations in Appendix D. The results are consistent.

Study 3: Qualitative evidence for why entrepreneurship education reduces recidivism

To better understand what about the PEP program helps participants once they are released, we conducted interviews and engaged in qualitative theory building. We interviewed 11 former participants of the PEP program about their experiences and conducted four interviews with program staff and leadership. Details of the interview procedures and analysis are provided in Appendix E.

Per above, there is evidence that starting a business plays a role in successfully integrating into the society for some participants. However, in our interviews both entrepreneurs and non-entrepreneurs explained why the program helped keep them from recidivating. Similarly, while our informants did discuss the resources and network that PEP offers as useful (Clough et al., 2019), they often talked about these as supporting benefits. In contrast, the explanation that emerges from our data is that entrepreneurship training reduces recidivism because it reinforces and makes tangible the character development training that participants receive first. That is, entrepreneurship training helps participants learn to set goals for, plan for, and take control of their lives, instead of just a business. Below, we detail each of these arguments.

Entrepreneurship as vision

The first finding that emerges from our data is that the entrepreneurship training changed the way participants thought about, and set goals for, the future. This is consistent with trained entrepreneurs being more adept at seeing opportunities to pursue (Baron & Ensley, 2006), however the scope here appears broader than evaluating businesses. The program allowed participants to imagine possibilities for themselves, and create visions of future-selves, which they could not have imagined in the past. In addition, it reinforced and complemented the character work provided in the first several months of the program, especially the focus on integrity.

PEP introduces the possibility of a new future as a “businessman” by leveraging successful former participants as role models of what is possible. Several participants told us stories of how seeing such a role model changed their belief about what was possible and who they could be. Whereas other entrepreneurs may get better at recognizing opportunities with training, PEP participants needed to be

able to recognize opportunities *and* believe that those opportunities were reachable for them. One described such an instance as follows:

“The guy that walked in was just sharp, walked in, in a nice suit with a briefcase, and I’m thinking, OK, this is, you know, very well, educated well put together guy... And then he showed us a video and in the video that he showed us, we saw that same guy in a white jumpsuit. And I was blown away. I’m like, like, right in front of me, I saw the transformation and I saw the opportunity for myself. Like, man, this guy could do it. Surely, I can do it too.” (NP1)

PEP used the entrepreneurship training, particularly the “businessman” identity to give the character lessons a purpose. As discussed earlier, PEP begins with their leadership academy. During this time, the program helps participants “wash the prison off” and promotes the ten driving values of the program. The participants linked the success of a possible future as a businessman to integrity and morals directly. For example, one summarized, *“To be a business owner and be a successful business owner, you have to have these qualities of accountability and integrity and excellence in pursuing all that.” (E2)*

This goal of becoming a “businessman” helped to change the way participants viewed their entire life, not just work. Our informants often cited “integrity” as one of values which continued to resonate with them long after completion. For instance, one participant highlighted the interconnected goals of succeeding in business and making a living with integrity:

“...[I]t’s like OK, if you cannot manage your own life, if you can’t keep your life on track, if you can’t be on a straight path, how can you, one, be a good employee for anybody else, or two, even own and operate and manage a small business and those people under you?” (E4)

Overall, while it is possible to provide character training without entrepreneurship training to help build a new identity (Rogers et al. 2017, Toubiana 2020, Zhang et al. 2021), our data also suggest that it may not have taken as well without entrepreneurship as a vision. Informants discussed how even if they wanted to do better once released, when things got challenging, they were drawn to reverting to the easier life they had always known. In these times, it was the promise of a better, more successful future that held them to the moral pathway. That is, they were focused on a potential “businessman” identity for the future, but that hinged on living with integrity first. On the other hand, teaching just entrepreneurship may not have led to such dramatic reductions in recidivism. As one participant said, *“You know yes, you can*

learn business. Yes, you can learn all how to do all types of things. But really in essence, if you're not changing your character, you're really just becoming a smarter criminal.” (E2)

Entrepreneurship as focused development

The second finding that emerges from our interview data is that the entrepreneurship training gave participants a focus on improving their practical skills and ability to plan. While business plans have been shunned by many entrepreneurship classes and training programs (Sarasvathy 2024), the analytic and planning skills in PEP became critical to participants’ lives outside of prison. As seasoned entrepreneurs rely on “scientific” skills of observation, questioning, and development of hypotheses, to find and build new opportunities (Leatherbee and Katila 2020, Camuffo et al. 2024), PEP participants honed their analytic skills and combined them with self-reflection to identify ways they could “level-up” to enact plans and reach their goals.

The participants all reported learning practical skills from the entrepreneurship training that were relevant for both starting their own businesses and getting or keeping gainful employment. Speaking (via toastmasters) and financial skills were among the most cited. One informant described this variety and how it helped them run their business, saying,

“We have everything from naming the business to marketing, you got to learn it all. It gave me enough smarts to have a conversation with somebody and not sound like I don't know what I'm talking about. So I can sit down with realtors and sound like a businessperson.” (E3)

However, of particular importance was the emphasis placed on planning or finding a path to achieve their goal. The business plan competition taught participants to map out how they would get from where they were at release to the envisioned goal of starting a business. This required analysis of competition, customer preferences, and a critical examination of their own and others’ plans. Participants learned to evaluate the risks of starting a business and skills needed to pursue it. This was critical for forming a plan of action. One participant explained,

“Knowing the processes of everything and doing your research before you dive into something because there's a bunch of business owners and people that start small businesses that just dive in because something intrigues them. So yeah, I want to be a part of that too. That without doing research and you know, within five years they fail because the preparation wasn't there. So when I went through the program was back in 2006. So from 2006 up until now has been preparation and waiting for the

opportunity to present itself and not just diving into something. So you know, doing the proper research and preparation before actually stepping forward [is] a huge thing [t]hat I learned also.” (E4)

Interestingly, even the participants who did not go on to start their own businesses talked about the planning and analytical skills as key lessons they took away from the program. One went as far as to say better planning combined with practical skills (e.g., understanding credit) were the reasons why he did not fall back into his old patterns of breaking the law:

“Well, I’ve always been able to get a good amount of money, ..., you know what I mean? I’ve always had plenty of money, I’ve always been able to pull that off. But then it would all like I wasn’t going to do nothing with it. I would just and then easy, easy come, easy go. And then you’d have like, 20 grand. And I couldn’t even find an apartment to rent because you can’t just show up with money and no job experience. But they taught me the importance of my credit getting right...[T]hen I just got the credit right. Before, I never had a plan. You have to have a plan, like some kind of plan. You got to be working towards some kind of a goal overtime and then even if the plans change, you’re still on track for something.” (E3)

Moreover, these planning skills honed their self-reflection and made them better at making decisions and creating plans for their lives, a lesson that also reinforced the initial character training. By combining self-reflection with focused development and planning in entrepreneurship training, they were able to better understand what skills they had, and what they needed to get where they wanted to be. As one participant put it,

“One thing that I remember a lot of like specifically, I remember them talking about is like, seeing, what do we need to work on? Where are we seeing the flaws? Where in our own life can I take an honest look?” (NE4)

This combination of self-reflection, practical skills, and planning allowed participants to avoid possible pitfalls in their new lives that could lead them back to prison. Overall, entrepreneurship once again reinforced and advanced the character training that participants received. It is well established that entrepreneurs who are trained to be more scientific in their decision making are better able to uncover flaws in their plans earlier and either avoid big losses or pivot to a new plan (Leatherbee and Katila 2020, Camuffo et al. 2024). This was true for PEP participants who considered starting a business. Informants often talked about not pursuing something because of too much competition or other factors. However, unlike entrepreneurs in other training programs, PEP participants had little to no experience planning in their lives and often did not have the needed skills to fall back on. The training to think more analytically,

plan, and reflect before acting gave them the ability to continue to improve themselves and their lives post-release in order to pursue their goals.

Entrepreneurship as control

The last finding that emerges from our data is that the training made participants see entrepreneurship as a metaphor for control. In a literal sense, entrepreneurship training gave the participants agency to start a business rather than be exposed to discrimination in the labor market (Dencker et al. 2021, Hwang and Phillips 2024). However, we again see the entrepreneurship training reinforce the character training that was provided first. In this case, the metaphor of entrepreneurship served to solidify lessons about accountability and taking ownership over one's life. The combination allowed participants to reenter the world with a new confidence that they could push through challenges both in business and their personal lives, instead of reverting to the easy way out.

The entrepreneurship training gave participants more ownership over their career decisions than they had experienced before. The difficulty of the entrepreneurship training (and the program as a whole) helped them to develop the *confidence* to take the business skills developed (as discussed above) and apply them. Participants described a feeling of *ownership* over those skills and choices to use them, repeatedly citing the slogan “*It's a hand up, not a handout.*” Ownership over their careers and confidence went hand in hand, for instance, “*OK, I'm gonna be the best whether it's washing dishes or whether it's running... being a CEO. Like PEP kind of gave you that confidence and I'm gonna go rock it. (NE4)*”

The ownership of their careers became a metaphor for ownership over their lives. As future businessmen, participants learn to take responsibility for their own actions and continue to pursue their goals. One former participant said of this metaphor,

“So, entrepreneurship is being your own boss, taking ownership of who you are, and leading your own life. It's not just about business. Entrepreneurship is being the decision maker, making your own decisions. That's something that really stuck out. Entrepreneurship wasn't about just running a business. It's about learning how to run your own lives.” (PE2)

In this way, the entrepreneurship training expanded the lessons of accountability participants were taught in character training. The reason for an emphasis on accountability is clear – if participants do not

take responsibility for the decisions and actions that put them in jail in the first place then they cannot grow beyond those. The program instills accountability and ownership in participants through strict rules:

“You gotta have your shirt tucked in. You cannot cuss. You have stuff to do from morning till night and you have to be accountable for any and everything you do wrong... If you do something wrong, they'll just give you like a dictionary and you like copy that by next week. The whole dictionary or whatever. It is something crazy like that... But if you really want to be in that program, you do it.” (E5)

This combination of accountability and ownership created resilience and a willingness to undertake challenges, both personally and professionally. A participant described this change in themselves saying,

“It truly changed me because I was the type of person before where if I saw a task [was] too hard I would give up. I'm like whatever. I'm like, man, that's too hard to even try, I'm not gonna try that no more. Forget that. It's too much work to go into it. And PEP helped me grow in that aspect.” (E4)

Overall, the entrepreneurship training solidified and expanded on the accountability training that had come before it by giving participants control over their employment outcomes. The idea that entrepreneurship training could give incarcerated individuals an alternative path to employment is not new (Hwang and Phillips 2024). However, we see that with that agency comes a confidence and resilience that participants feel they can handle setbacks both in their employment and their lives more generally. Where other entrepreneurial training programs might teach participants how to experiment and pivot (Camuffo et al. 2024), the PEP participants first needed to learn that it was OK to fail and that they could succeed in doing hard things because they had ownership over their choices.

DISCUSSION

This study seeks to contribute primarily to growing research on entrepreneurship training programs, particularly for disadvantaged populations (Dutt et al. 2016, Assenova 2020, Hallen et al. 2020). While recent research has uncovered numerous ways that entrepreneurship training programs may help nascent entrepreneurs, findings on the effectiveness of such training for entrepreneurs without high-growth objectives and a high baseline of human capital are mixed. Our findings suggest that training programs may make a substantive influence for incarcerated individuals by reducing recidivism and

increasing the ability to reintegrate into society. In addition, we find a negative association between entrepreneurial entry and recidivism, and the association is stronger than that for individuals who were employed and those who were not employed upon release. This finding underlines the potential for entrepreneurship to play a significant role in addressing structural barriers (e.g., Rindova et al. 2009, Hwang and Phillips 2024, Yang et al. 2024).

In addition, our focus on recidivism as a potential outcome of entrepreneurial training extends prior work by exploring the impact of entrepreneurship and entrepreneurial training beyond profit maximization or growth. The study highlights that outcomes of interest may be more varied and entrepreneurial training and the resultant activity may have wider repercussions for individuals and society beyond venture success. Our work joins an emerging literature that explores the relationship between entrepreneurial activities for communities and the society at large (Fehder and Hochberg 2014, Kacperczyk and Rocha 2021, Kwon and Sorenson 2023, Hwang and Phillips 2024). It also provides an additional role for entrepreneurship training as a supporting institution for disadvantaged populations that may lack them (Dutt et al. 2016).

While the results of the archival data analysis underline the potential of entrepreneurship to address the problem of recidivism, our qualitative interviews highlight additional insights about the training program that might be relevant for future research and practice. We find that the program helps participants not only by providing technical and analytical skills, but also by reinforcing the lessons from the prior character training. We found that entrepreneurship played three distinct roles in preparing formerly incarcerated individuals for release: By acting as a vision, an opportunity for focused development, and as a metaphor for control. In this way, the program gave participants a goal, the tools to find a path to achieve that goal, and a way to assume responsibility for their past and future. The program graduates emphasized these particular lessons, beyond entrepreneurial skills, in explaining how the program helped them succeed after release. It appeared that the entrepreneurship and character development lessons complemented each other such that the delivery of one without the other would not be as effective. This finding is consistent with prior findings in the literature on the complementarity of

behavioral interventions and business training, particularly for disadvantaged entrepreneurs (Patzelt et al. 2014, Campos et al. 2017, Dimitriadis and Koning 2022).

To be sure, the study has several important limitations. First, the data is from a single program and we cannot rule out the possibility that the results do not reflect the idiosyncratic characteristics of the site, program, delivery, and participants. Second, we have limited data on incarcerated individuals that prevent us from carefully controlling for individual characteristics. Obtaining more detailed data on incarcerated individuals is not possible due to strict confidentiality requirements. At the same time, we obtain convergent support and strong results across the three studies we included in this paper. Future work can tease out the factors that influence the effectiveness of training programs by examining different populations that face differential levels of barriers, which may then inform the design of training programs. Finally, we do not expect entrepreneurship to be the panacea to the problems of reintegration and recidivism for all. It is important to underline that those who apply, get accepted, and complete the program is a selected subset of all incarcerated individuals. Instead, the correct interpretation of the results is that entrepreneurship training can be a path toward reintegration for a subset of incarcerated individuals who are willing and able to pursue this path. In spite of the limitations, this study aims to contribute by examining whether and how entrepreneurship training may help address societal challenges, contribute solutions to outcomes of importance to communities, and distribute the benefits of entrepreneurship more evenly across the society.

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TABLES

Table 1. Balance Table

	Before matching					After matching				
	Untrained (n=1,540)		Trained (n=969)			Untrained (n=1,509)		Trained (n=952)		
	Mean	Std. dev.	Mean	Std. dev.	t-test	Mean	Std. dev.	Mean	Std. dev.	t-test
White	0.366	0.482	0.334	0.472	1.592 (p=0.111)	0.371	0.483	0.339	0.474	1.603 (p=0.109)
Black	0.351	0.477	0.311	0.463	2.065 (p=0.039)	0.345	0.475	0.316	0.465	1.490 (p=0.136)
Hispanic	0.272	0.446	0.324	0.468	-2.754 (p=0.006)	0.277	0.448	0.329	0.470	-2.740 (p=0.006)
Age	34.068	8.079	34.325	7.504	-0.795 (p=0.426)	34.056	7.88	34.282	7.447	-0.706 (0.480)
Release Year	2013.498	1.934	2013.128	2.274	4.356 (p=0.000)	2013.524	1.927	2013.15	2.264	4.369 (0.000)
	Multivariate L1 distance: 0.346					Multivariate L1 distance: 0.265				

Table 2: Comparison of 3-year Recidivism for Trained and Untrained Groups

	3-year recidivism			Total recidivism		
Recidivated	Untrained	Trained	Total	Untrained	Trained	Total
No	1,199	885	2,084	1,035	814	1,849
Yes	341	84	425	505	155	660
Total	1,540	969	2,509	1,540	969	2,509
Rate of recidivism (%)	22.14	8.67		32.79	15.99	

Table 3: Logistic Regression Models of 3-Year and Total Recidivism

	Unbalanced sample				Matched sample with CEM weights			
	(1)	(2)	(3)	(4)	(3)	(4)	(5)	(6)
	3-year recidivism	Total recidivism	3-year recidivism	Total recidivism	3-year recidivism	Total recidivism	3-year recidivism	Total recidivism
Trained	-1.082***	-0.951***	-1.072***	-0.943***	-1.065***	-0.920***	-1.070***	-0.929***
	(0.131)	(0.106)	(0.132)	(0.107)	(0.138)	(0.112)	(0.138)	(0.113)
Black			-0.318**	-0.272**			-0.203	-0.188
			(0.132)	(0.112)			(0.146)	(0.124)
Hispanic			-0.369***	-0.375***			-0.272	-0.313**
			(0.139)	(0.119)			(0.177)	(0.147)
Other			-0.573	-0.777*			-0.815	-1.017
			(0.478)	(0.429)			(0.680)	(0.621)
Age 25-34			0.0452	0.0888			0.258	0.172
			(0.170)	(0.146)			(0.205)	(0.173)
Age 35-44			-0.235	-0.261			0.000530	-0.133
			(0.185)	(0.160)			(0.214)	(0.182)
Age 45-54			-0.500**	-0.529**			-0.224	-0.419*
			(0.246)	(0.210)			(0.299)	(0.252)
Age 55+			-0.521	-1.234			-0.492	-1.473
			(0.805)	(0.780)			(1.249)	(1.235)
Constant	-1.232***	-0.857***	-0.981***	-0.620**	-1.634***	-1.111***	-1.673***	-1.039***
	(0.271)	(0.237)	(0.317)	(0.273)	(0.316)	(0.293)	(0.369)	(0.334)
Year Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,509	2,509	2,509	2,509	2,461	2,461	2,461	2,461
Pseudo R-sq	0.0394	0.0421	0.0472	0.0529	0.00877	0.0210	0.0377	0.0423
Log likelihood	-1096	-1385	-1088	-1369	-1098	-1383	-1066	-1353

Release year dummies included in all models. Clustered standard errors in parentheses. ***p<0.01, **p<0.05, *p<0.1

Table 4. Comparison of recidivism rates by post-release status

3-year recidivism				
	Obs.	Mean	Std. dev.	t-test of mean difference (vs. unemployed)
Entrepreneur	185	0.022	0.011	0.655 (p=0.513)
Employed	239	0.054	0.015	-0.696 (p=0.487)
Unemployed	85	0.035	0.020	
Total recidivism				
	Obs.	Mean	Std. dev.	t-test of mean difference (vs. unemployed)
Entrepreneur	185	0.043	0.015	3.721 (p=0.0002)
Employed	239	0.084	0.018	2.380 (p=0.018)
Unemployed	85	0.176	0.042	

Table 5. Linear Probability Models of Entrepreneurship, Employment, and Recidivism Rates

	(1)	(2)	(3)	(4)	(5)	(6)
	3-year Recidivism	Total recidivism	3-year Recidivism	Total recidivism	3-year Recidivism	Total recidivism
Entrepreneur	-0.029*	-0.067***			-0.025	-0.152***
	(0.016)	(0.023)			(0.023)	(0.045)
Employed			0.023	-0.009	0.0056	-0.115**
			(0.017)	(0.024)	(0.024)	(0.046)
Black	-0.0016	-0.021	-0.001	-0.019	-0.002	-0.022
	(0.023)	(0.032)	(0.023)	(0.032)	(0.023)	(0.032)
Hispanic	-0.012	-0.014	-0.010	-0.006	-0.012	-0.014
	(0.022)	(0.031)	(0.022)	(0.032)	(0.022)	(0.031)
Other	0.0095	0.068	0.007	0.064	0.009	0.079
	(0.050)	(0.081)	(0.050)	(0.081)	(0.051)	(0.080)
Age 25-34	0.038***	-0.004	0.038**	0.0002	0.038**	0.002
	(0.015)	(0.048)	(0.015)	(0.049)	(0.015)	(0.047)
Age 35-44	0.039**	-0.002	0.039**	0.001	0.040**	0.005
	(0.017)	(0.049)	(0.017)	(0.050)	(0.017)	(0.047)
Age 45-54	0.024	0.010	0.023	0.014	0.0238	0.017
	(0.031)	(0.064)	(0.032)	(0.065)	(0.032)	(0.062)
Age 55+	0.017	-0.062	0.019	-0.055	0.017	-0.062
	(0.018)	(0.052)	(0.018)	(0.050)	(0.018)	(0.059)
Constant	0.007	0.089	-0.009	0.059	0.005	0.146**
	(0.028)	(0.058)	(0.027)	(0.056)	(0.034)	(0.066)
Year Dummies	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.040	0.027	0.038	0.014	0.040	0.046
Log likelihood	119.2	-65.30	118.8	-68.62	119.3	-60.09
Linear probability models. N=505. Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1						

FIGURES

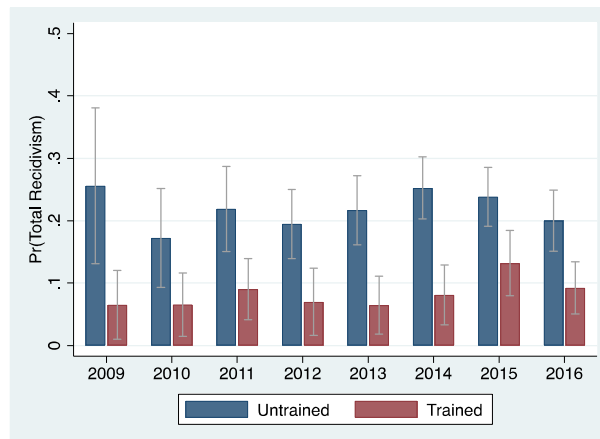


Figure 1. Three-year recidivism rates by release year

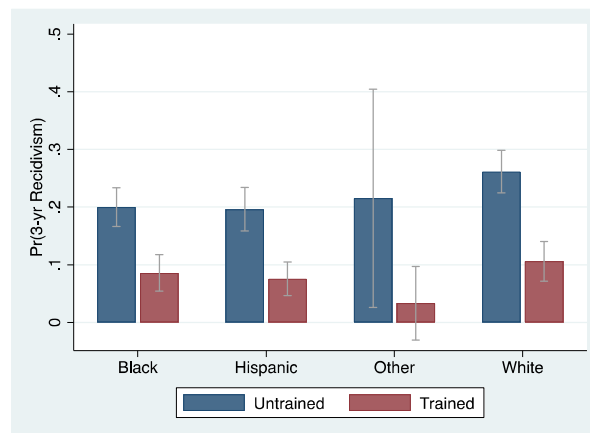


Figure 2. Three-year recidivism rates with and without training by race

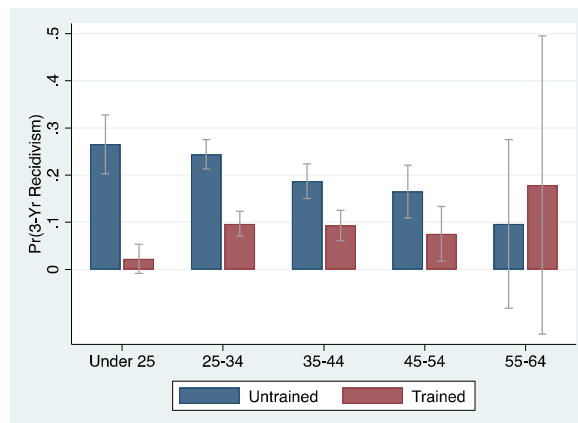


Figure 3. Three-year recidivism rates with and without training, by age group

APPENDICES

Appendix A: Comparison of Entrepreneurship Training Programs in the US

	 PEP	 Defy Ventures	 BEST	 LEAP	 LIFE	 OPEP	 Resilience Education	 The Last Mile	 Entrepreneurs Inmates to Entrepreneurs	 Project ReMADE
Program Type	In-prison and post-release		In-prison						Post-release	
Location	Texas	CA, CO, CT, NE, NJ, NY	Tennessee	Florida	Oregon	Ohio	Virginia	California, Indiana	North Carolina	California
Year Established	2004	2010	2014	2009	2007	2017	2011	2010	2008	2011
Annual Participants	>1,000	>1,000	15-30	20-40	20-40	15-30	10-20	Not reported	Not reported	<10
In-Prison Program Timing	Up to 9 months	Up to 14 months	6 months	8 months	7 months	3 months	9 months	6 months	Not applicable	Not applicable
Post-Release Services Timing	Indefinite (re-entry, education)	Indefinite (re-entry, education)	Indefinite (education)	Not reported	Indefinite (re-entry)	Not reported	Not reported	Not reported	2 months (education)	3 months (education)

Source: ICIC Impact Analysis, 2018.

Appendix B: Program Process Flow – Recruitment to Release

The graphic below provides an illustrative example of a typical quarterly recruiting cycle for PEP. The cycle starts with the mailing of an average of 2,000 recruiting letters to qualifying incarcerated individuals in the Texas prison system each quarter. An individual must be within three years of release, have or be in the process of attaining a high school diploma or GED, and cannot be an active gang member or sexual offender. PEP receives approximately 600 responses out of the 2,000 letters sent. Those who are interested receive a 20-page application in addition to a study packet and a 50-question test. The application gathers information on the characteristics of the applicants (such as demographics, type of charge, duration of incarceration, criminal, drug, alcohol, family, schooling, and gang history), asks applicants to submit a testimonial essay on why they are a good fit as well as a walk-through of their life timelines. Application packages are scored based on a proprietary SWOT ranking system to evaluate the perceived probability of completing the training and embodying the values of the program upon release. PEP staff use their own judgment based on prior experience to categorize life factors as either positive (strengths and opportunities) or negative (weaknesses and threats). For example, life as a former drug dealer is scored as a strength due to the entrepreneurial capacity displayed by such former activities. An example of a weakness is if the incarcerated individual's family uses drugs in the home, which is believed to place the incarcerated individual at high risk of engaging in criminal activity upon release. Thus, the SWOT analysis helps select those that are most likely to become successful entrepreneurs conditional upon the incarcerated individual showing initial interest in entrepreneurship.

Candidates who receive high scores on the test and SWOT assessment are interviewed by the program staff, which in this example reduces the pool to about 400 candidates. Interviews are conducted in person when possible, and on a detailed paper application for individuals located in prison facilities located too far from the PEP headquarters to make in-person meetings feasible. Approximately half of the interviews are in person and half are on paper. The program notes no differences in historical results based on interview type, as the SWOT analysis is believed to perform the primary selection function. The only reason some applicants are interviewed in person is if the originating prison is close enough to the

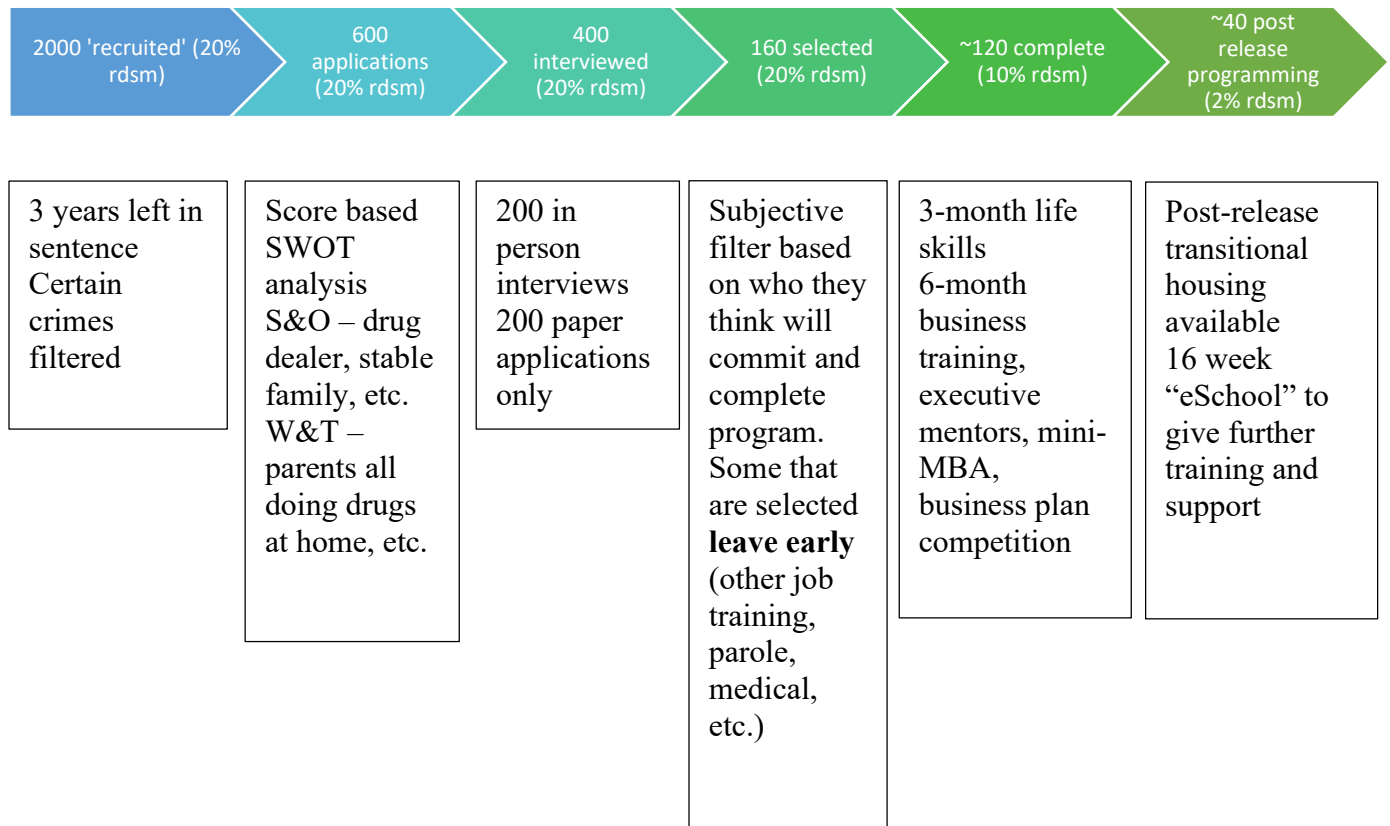
PEP headquarters, with the more distant incarcerated individuals mailing in paper applications. The interviews are used to gather additional data which is used to update the SWOT scoring analysis. Approximately 160 individuals are selected to receive training based on the interviews. In 2017, 54% of PEP graduates surveyed had started or completed vocational classes, trade school or college before prison, and 13% had previous entrepreneurship experience (ICIC 2018).

These individuals are physically relocated to one of two in-person training prisons – one in Houston and the other in Dallas. These program participants receive 3 months of life skills training designed to promote a growth mindset and teach skills around conflict resolution, social etiquette, drug and alcohol abuse and dating/marriage, centering around the program’s Ten Driving Values ⁴. Following this phase, the candidates receive a 6-month ‘mini-MBA’ where they learn basics of accounting, marketing, finance, operations and other core business skills. The program follows a textbook titled “Entrepreneurship: A Small Business Approach” (Bamford and Bruton 2011) in addition to financial literacy courses, business etiquette, and Toastmasters public speaking exercises. During this phase of the program, all participants prepare a business plan around an entrepreneurial idea. Executives from Houston and Dallas based firms volunteer as mentors to aid the participants in honing their business plans, and MBA students from Baylor university volunteer to conduct detailed research to validate their approaches (as the incarcerated individuals are not able to engage in extensive online research due to facility limitations).

The program culminates with a business plan competition and graduation where family and friends often come to celebrate. Not all program participants graduate, with some leaving the program due to disciplinary issues. In this example, 120 participants reach graduation. Finally, an optional “eSchool” and reintegration program is available to those that are released and wish for additional assistance. The eSchool provides further in-person entrepreneurial training and the reintegration program provides

⁴ <https://www.pep.org/ten-driving-values/>

transitional housing, financial assistance and to help these individuals start their businesses. Roughly 40 of the 120 graduates will choose to receive this additional help.



Sources:

Interviews with program staff, PEP graduate survey,

https://icic.org/wp-content/uploads/2018/07/ICIC_PEP-Impact-Analysis_final_post.pdf

<https://www.aei.org/wp-content/uploads/2016/12/Prison-Entrepreneurship-Program.pdf>

Appendix C. Details of the parole process

In our analyses, we examined individuals who were admitted to the program but did not attend, since they received parole within the window between program admission and training start, as a comparison set to individuals who were admitted and received training. If those individuals who were released on parole were systematically different in ways that are unobservable to us and would increase their recidivism rates, our inferences may be biased. In this Appendix we provide more background on the parole process to justify the assumption that the two subsamples are comparable.

Parole refers to the release of an offender to serve the remainder of their sentence under supervision in the community, subject to conditions. Parole requirements are legislated at the state level and each offender is evaluated for parole approval by a parole panel. Parole is discretionary and the parole board is not obligated to approve parole for an individual at any time. In order to be approved for parole release, an offender must serve sufficient time as required by state law (for most crimes, one fourth of their sentence), must not pose a risk to public safety, and must meet work, program participation, and behavior standards. For many offenders, participating in work and self-improvement programs count as ‘good time’ and may be added to calendar time served in calculating their eligibility for parole.

Parole Review Process

An offender receives a time calculation sheet showing their initial parole eligibility date, provided by the TDCJ Classification and Records Department. The date on the sheet is subject to revision based on good time earned or lost. The parole review process begins several months before an offender’s parole eligibility review date. Once the offender provides the appropriate documentation, the case is assigned to an institutional parole officer (IPO), who prepares a case summary. The case is then reviewed by a parole panel that includes three panelists. Panelists vote sequentially and two out of three must vote in favor of parole for approval. Panelists assess cases based on the probability of parole success. The factors that the IPO and panelists consider include the seriousness of the offense, sentence length and amount of time served, prior criminal behavior and substance abuse history, institutional history (Education, vocational, and certified on-the-job training programs completed during the present incarceration), and age.

The parole panel has a variety of voting options, including release when eligible, release on a specified future date, transfer to a TDCJ rehabilitation program, deny parole and set time for next parole consideration, and deny parole with no subsequent review. Parole approval rates in Texas were 34% in 2016.

Parole Requirements

Offenders released on parole are required to abide by certain rules and conditions, such as reporting as instructed to their parole officer, obeying all municipal, country, state, and federal laws, not owning, possessing, selling, or controlling any firearm or prohibited weapon, taking drug tests as required, obtaining permission of the parole officer for travel and residence change, and appearing at any required hearings or proceedings.

Several programs are available in the state of Texas to assist individuals released on parole, such as District Reentry Centers for high-risk offenders, substance abuse treatment centers, education assistance, and employment assistance. If an offender violates their terms of release, sanctions vary from a letter of reprimand from the Parole Division to an arrest warrant (called a “blue warrant”) and revocation of parole, depending on the severity of the violation. An offender whose parole has been revoked must serve the rest of their sentence. In 2015, the 3-year reincarceration rate among those starting parole supervision was 20.2%.

Based on this background, we can address several possibilities that could threaten the validity of using those who were accepted to the program but were released on parole after program acceptance but before training as a counterfactual to those who were accepted to the program and received training. First, we need to understand whether those who received parole were more likely to recidivate compared to those who did not receive parole. This appears unlikely, since individuals are carefully screened by the parole board for their likelihood of successfully completing parole (*i.e.*, not committing another crime during their parole supervision period). As a result, we would expect them to be less, not more, likely to commit a crime compared to their counterparts. Indeed, according to the statistics above, the likelihood of reincarceration within 3 years was 20.2% In contrast, the 3-year recidivism rate for those released from

states prisons and jails in the same period in Texas was 25.6%. This suggests that individuals who were released on parole were not more likely to recidivate than the general prison population.

At the same time, it is possible that the individuals who received parole after getting admitted to the program have different characteristics than the broader group of parolees. For instance, if individuals can accurately predict their likelihood of receiving parole, those who are more likely to receive parole may not apply to the program. If this were true, those who applied to the program may be the ones who were less likely to receive parole, and more likely to recidivate later. This seems unlikely for several reasons. First, given the extensive parole review, it is unlikely for a large number of individuals to accurately predict the outcome of parole review and timing of release after a favorable review. Second, the PEP program rigorously screens all applicants (see Appendix B on program selection criteria) and this evaluation is independent from that of the parole board. It is very unlikely for PEP to select individuals who were more likely to receive parole but were more likely to recidivate. In fact, many criteria for program admission overlap with parole approval criteria. This is why the rate of parole release is much higher in the group who has been accepted to attend the program (1540 out of 2741 individuals in the sample received parole, or 56%) compared to the one in the general Texas prison population (34% of all cases considered). Indeed, the 3-year reincarceration rates for those who were released on parole supervision with or without program acceptance are consistent (22.1% vs. 20.2%), supporting this argument.

Sources

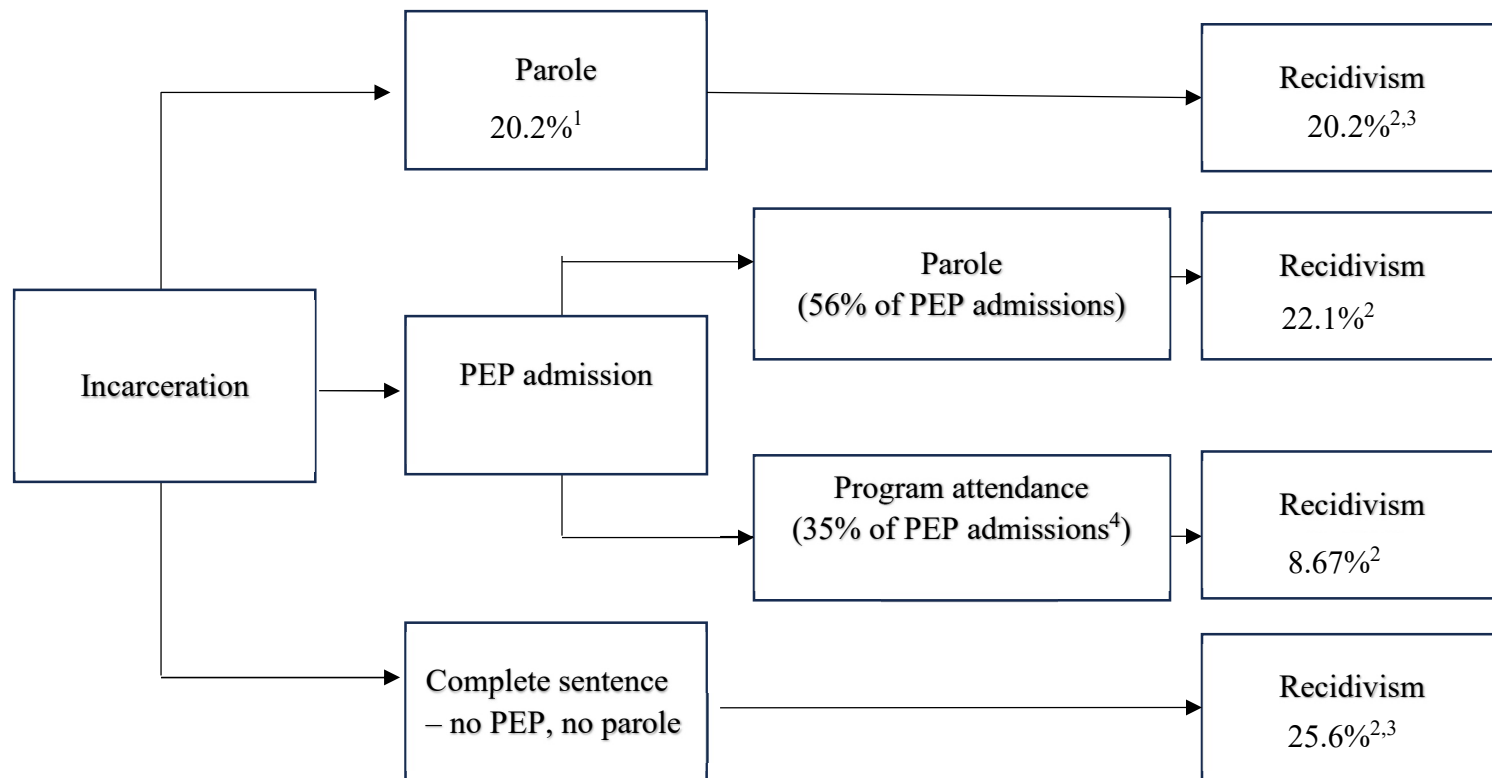
https://www.tdcj.texas.gov/bpp/publications/PIT_English.pdf

https://www.tdcj.texas.gov/bpp/publications/FY_2020_Annual_Statistical_Report.pdf

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https://www.tdcj.texas.gov/documents/rid/RID_Reentry_Biennial_Report_09_2022.pdf

https://www.lbb.texas.gov/documents/publications/policy_report/4914_recivism_revocation_rates_jan2019.pdf



¹Based on statistics provided by the Texas Board of Pardons and Paroles

² Percentages show recidivism rates within a given subgroup, for instance, recidivism rates for those who received parole in the state of Texas were 20.2%. For program attendees, rates are based on analyses reported in the manuscript.

³https://www.lbb.texas.gov/documents/publications/policy_report/4914_recidivism_revocation_rates_jan2019.pdf

⁴ Percentages of PEP admissions do not add up to 100% because of those who dropped out due to health discharge

Appendix D. Supplementary models

Table D1a: Descriptive statistics of accepted and trained individuals

Age Group	Black	Hispanic	Other	White	Total	% of Total
Under 25	22	37	3	25	87	8.98%
25-34	156	169	18	141	484	49.95%
35-44	102	88	9	112	311	32.09%
45-54	21	18	0	42	81	8.36%
55-64	0	2	0	4	6	0.62%
Total	301	314	30	324	969	
% of Total	31.06%	32.40%	3.10%	33.44%		

Table D1b: Descriptive statistics of accepted but not trained individuals

Age Group	Black	Hispanic	Other	White	Total	% of Total
Under 25	67	68	4	56	195	12.66%
25-34	283	197	11	244	735	47.73%
35-44	142	132	0	157	431	27.99%
45-54	44	22	1	101	168	10.91%
55-64	4	1	1	5	11	0.71%
Total	540	420	17	563	1,540	
% of Total	35.06%	27.27%	1.10%	36.56%		

Table D2: Linear probability and Cox proportional hazard models of recidivism

	1	2	3
	3-year recidivism	Total recidivism	Total Recidivism
	LPM	LPM	Cox proportional hazard
Trained	-0.130*** (0.014)	-0.164*** (0.017)	-0.844*** (0.0937)
Black	-0.045** (0.019)	-0.052** (0.021)	-0.243*** (0.0925)
Hispanic	-0.051*** (0.019)	-0.069*** (0.022)	-0.318*** (0.0999)
Asian	-0.068 (0.045)	-0.122** (0.054)	-0.656* 0.0570
Age – 25-34	0.008 (0.025)	0.015 (0.029)	(0.122) -0.230*
Age – 35-44	-0.031 (0.026)	-0.048 (0.030)	(0.136) -0.459**
Age – 45-54	-0.065** (0.032)	-0.096*** (0.037)	(0.179) -1.084
Age 55+	-0.065 (0.084)	-0.195** (0.084)	(0.717) 0.0570
Constant	0.257*** (0.041)	0.349*** (0.047)	
Release year dummies	Yes	Yes	Yes
Observations	2,509	2,509	2,504
R-Squared	0.040	0.057	
Log Likelihood			-4944

Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

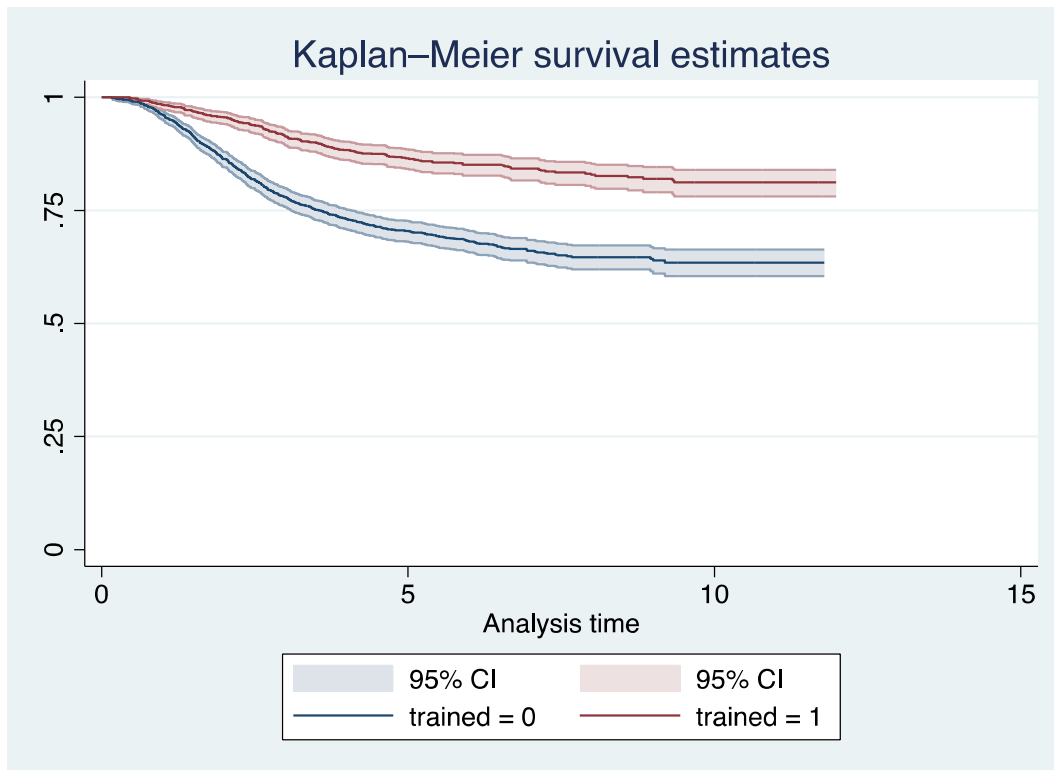


Figure D1: Survival Analysis by Training Status

Table D3: Validation with Multiple Matching Methods

Model:	1	2	3	4	5	6
Dependent Variable:	3-Year Recidivism	Total Recidivism	3-Year Recidivism	Total Recidivism	3-Year Recidivism	Total Recidivism
Trained	-0.139*** (0.016)	-0.176*** (0.020)	-0.137*** (0.017)	-0.173*** (0.020)	-0.133*** (0.015)	-0.163*** (0.018)
Matching Method	Exact	Exact	Nearest Neighbor	Nearest Neighbor	Propensity Score Matching	Propensity Score Matching

N=2,509. Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D4. Comparison of parole vs. discharge

		3-year recidivism			Total recidivism		
	n	Mean	St. Dev.	t-test	Mean	St. Dev.	t-test
Parole	1540	0.221	0.415		0.328	0.469	
Discharge	232	0.198	0.3999	-0.795 (p=0.426)	0.306	0.461	-0.663 (p=0.597)

Table D5. Logistic regression models of recidivism (comparison set: individuals who were admitted to PEP but released on medical discharge)

	1	2
	3-year Recidivism	Total recidivism
Trained	-1.001***	-0.923***
	(0.205)	(0.175)
Black	-0.260	-0.306
	(0.236)	(0.188)
Hispanic	-0.227	-0.143
	(0.238)	(0.189)
Other	-1.168	-0.649
	(1.050)	(0.629)
Age 25-34	0.868**	0.284
	(0.386)	(0.265)
Age 35-44	0.823**	0.241
	(0.399)	(0.279)
Age 45-54	0.791*	0.308
	(0.479)	(0.341)
Age 55+	1.540	0.028
	(1.117)	(1.108)
Constant	-2.438***	-1.070**
	(0.610)	(0.423)
Release year dummies	Yes	Yes
Observations	1,201	1,201
Pseudo R-sq.	0.045	0.037
Log likelihood	-393.2	-559.5

Robust standard errors in parentheses. ***p<0.01, **p<0.05, *p<0.1

Table D6. Tabulation of recidivism outcomes by completion of Reintegration Program (RP)

	n	3-year Recidivism			Total Recidivism		
		Mean	Std. dev.	t-test	Mean	Std. dev.	t-test
No RP	609	0.103	0.012		0.175	0.154	
RP	360	0.058	0.012	2.416 (p=0.016)	0.133	0.018	1.739 (p=0.082)

Table D7. Logistic models of Reintegration Program (RP) completion and recidivism

	1	2
	3-year Recidivism	Total recidivism
Completed RP	-1.509***	-1.211***
	(0.237)	(0.171)
Did not complete RP	-0.875***	-0.804***
	(0.148)	(0.123)
Black	-0.326**	-0.279**
	(0.132)	(0.112)
Hispanic	-0.364***	-0.373***
	(0.139)	(0.120)
Other	-0.585	-0.783*
	(0.477)	(0.426)
Age 25-34	0.0498	0.0949
	(0.170)	(0.146)
Age 35-44	-0.234	-0.258
	(0.185)	(0.160)
Age 45-54	-0.506**	-0.534**
	(0.246)	(0.210)
Age 55+	-0.510	-1.231
	(0.817)	(0.789)
Constant	-0.983***	-0.623**
	(0.319)	(0.273)
Release year dummies	Yes	Yes
Observations	2,509	2,509
Pseudo R-sq.	0.049	0.050
Log likelihood	-1084.421	-1084.421

Robust standard errors in parentheses. ***p<0.01, **p<0.05, *p<0.1

Table D8: Linear probability models of interaction effects

	(1)	(2)	(3)	(4)
	3-year Recidivism	Total recidivism	3-year Recidivism	Total recidivism
Trained	-0.150*** (0.025)	-0.171*** (0.030)	-0.242*** (0.036)	-0.264*** (0.048)
Black	-0.058** (0.025)	-0.047* (0.028)	-0.046** (0.019)	-0.053** (0.021)
Hispanic	-0.062** (0.027)	-0.083*** (0.030)	-0.050*** (0.019)	-0.068*** (0.022)
Other	-0.039 (0.104)	-0.147 (0.104)	-0.067 (0.045)	-0.119** (0.055)
Age 25-34	0.007 (0.025)	0.015 (0.029)	-0.022 (0.035)	-0.000 (0.039)
Age 35-44	-0.030 (0.026)	-0.048 (0.030)	-0.081** (0.037)	-0.102** (0.041)
Age 45-54	-0.065** (0.032)	-0.096*** (0.037)	-0.114*** (0.043)	-0.164*** (0.048)
Age 55+	-0.066 (0.085)	-0.196** (0.084)	-0.169* (0.094)	-0.292*** (0.093)
Trained * Black	0.034 (0.034)	-0.012 (0.041)		
Trained * Hispanic	0.028 (0.035)	0.033 (0.043)		
Trained * Other	-0.037 (0.111)	0.042 (0.121)		
Trained * Age 25-34			0.100** (0.041)	0.061 (0.054)
Trained * Age 35-44			0.150*** (0.044)	0.156*** (0.056)
Trained * Age 45-54			0.158*** (0.055)	0.217*** (0.071)
Trained * Age 55+			0.312* (0.177)	0.286 (0.178)
Constant	0.266*** (0.043)	0.352*** (0.048)	0.295*** (0.046)	0.386*** (0.052)
Release year dummies	Yes	Yes	Yes	Yes
R-squared	0.041	0.058	0.044	0.062
N=2,380. Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1				

Table D9. Descriptive statistics for survey respondents (n=505)

Variable	Mean	Std. dev.	Min	Max
3-year recidivism	0.04	0.19	0	1
Combined recidivism	0.08	0.28	0	1
Started business	0.36	0.48	0	1
Employed	0.47	0.50	0	1
White	0.32	0.47	0	1
Black	0.30	0.46	0	1
Hispanic	0.33	0.47	0	1
Age	34.68	7.69	19.4	62.80556
Release year	2013.22	2.19	2009	2016

Sample selection bias and endogeneity in survey-based analyses

The results of the analyses investigating the relationship between post-release entrepreneurship and recidivism based on the survey of the graduates has two important limitations. First, we only have entrepreneurship and employment data for those who have filled the survey, which could cause sample selection bias. The demographic characteristics of survey respondents is consistent with those who completed the program in the observational data, which alleviates the concern. In addition, we reported earlier that even if the entrepreneurial entry rate among 464 non-respondents was zero, the rate of entrepreneurship would be $185/969=19.1\%$, which is still higher than the reported rates in prior work.

Nevertheless, in order to address this issue, we ran linear regression models adjusting for nonresponse using survey probability weights, using `svy` function in Stata. We first calculated predicted response rates for each individual based on race, age group, and release year. We then ran survey regression models estimating the relationship between recidivism and entrepreneurship, implementing probability weights obtained in the previous stage. As reported in Table D10, the results are consistent and show a negative relationship between entrepreneurship and recidivism. Next, we ran probit models with sample selection (Van de Ven and Van Pragg 1981), which predict the likelihood of filling the survey as a function of race, age, and release year. We used a dummy variable that identifies whether an

individual is released to a family member (“warm release”) to fulfill the exclusion restriction, since those who were released to family members were less likely to fill the survey and no more likely to recidivate. The results, reported in Table D11, show that the relationship between entrepreneurship and total recidivism is negative and significant, while that between entrepreneurship and 3-year recidivism is negative though not significant. These results are consistent with the ones reported in Table 5.

The second issue is that those who choose entrepreneurship may be different in unobserved ways from those who did not choose entrepreneurship as a career path, causing an omitted variable bias. In order to address this concern, we ran single equation 2-stage instrumental variable regressions in which we predicted the likelihood of selecting into entrepreneurship in the first stage (Wooldridge, 2010)⁵. As an instrument for entrepreneurship, we used survey respondents’ answer to the multiple response question, “Which parts of the program were useful for you?” We coded the instrument as 1 if an individual chose entrepreneurship training as the most useful part of the program, and 0 otherwise. This variable captures an individual’s interest in entrepreneurship but is not related to the likelihood to recidivate for any other reason than becoming an entrepreneur, satisfying the exclusion criterion⁶. Thirty four percent of individuals who have become entrepreneurs have indicated entrepreneurship training as the most useful part of the program, while 20% of other individuals did. As reported in Table D12, the results are consistent with those reported earlier. The relationship between entrepreneurship and both 3-year and total recidivism is negative and significant. Together, these results provide supporting evidence of entrepreneurship as a potential path toward long-term reintegration.

⁵ We obtained robust results with probit-two stage least squares (probit-2SLS) models..

⁶ In alternate models, we also coded the variable as [1/number of other program components marked as most useful] for those who chose entrepreneurship training as useful, and 0 for others, generating a continuous instrument. The results were similar.

Table D10. Survey regression results with sample weights

	(1)	(2)
	3-Year Recidivism	Total Recidivism
Trained	-0.030*	-0.065***
	(0.016)	(0.022)
Black	-0.006	-0.027
	(0.022)	(0.031)
Hispanic	-0.011	-0.014
	(0.022)	(0.031)
Other	0.010	0.064
	(0.050)	(0.079)
Age 25-34		0.004
		(0.027)
Age 35-44		0.0218
		(0.052)
Age 45-54		-0.056**
		(0.028)
Age 55+		0.007
		(0.049)
Constant	0.036	0.080**
	(0.046)	(0.039)
Observations	505	505
R-squared	0.039	0.027
Standard errors in parentheses		
*** p<0.01, ** p<0.05, * p<0.1		

Table D11. Probit models with sample selection predicting recidivism

	(1)	(2)	(3)	(4)
	3-year recidivism		Total recidivism	
	Survey	Recidivism	Survey	Recidivism
Entrepreneur		-0.181		-0.383**
		(0.136)		(0.191)
Black	-0.037	0.038	-0.041	-0.126
	(0.103)	(0.136)	(0.103)	(0.172)
Hispanic	0.094	-0.133	0.095	-0.141
	(0.102)	(0.142)	(0.102)	(0.154)
Other	0.683**	-0.288	0.645**	-0.0530
	(0.271)	(0.311)	(0.257)	(0.401)
Age	0.0042	-0.004	0.0044	-0.006
	(0.006)	(0.007)	(0.006)	(0.009)
Warm release	-0.346***		-0.350***	
	(0.099)		(0.093)	
Rho		-1.952**		-1.056
		(0.814)		(0.691)
Constant	-0.115	-0.036	-0.120	-0.242
	(0.244)	(0.429)	(0.245)	(0.670)
Release year dummies	Yes	Yes	Yes	Yes
chi2	19.50	19.50	10.77	10.77
Log likelihood	-721.4	-721.4	-789.6	-789.6

n=969. Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table D12. Single Equation 2-Stage Instrumental Variable Regression of Recidivism

	(1)	(2)
	3-year Recidivism	Total recidivism
Entrepreneur	-0.233**	-0.509***
	(0.110)	(0.179)
Black	-0.0301	-0.130
	(0.0505)	(0.0959)
White	-0.0213	-0.0934
	(0.0515)	(0.0958)
Hispanic	-0.0545	-0.154
	(0.0536)	(0.100)
Age 25-34	0.00522	0.00955
	(0.0226)	(0.0349)
Age 35-44	-0.0143	0.0128
	(0.0362)	(0.0600)
Age 45-54	-0.0319	-0.0819
	(0.0699)	(0.137)
Age 55+	-0.0292	0.0234
	(0.0241)	(0.0584)
Constant	0.155*	0.369***
	(0.0882)	(0.139)
Observations	505	505
chi2	16.03	11.35
F-statistic	12.198 (p = 0.0005)	12.198 (p = 0.0005)
Robust standard errors in parentheses		
*** p<0.01, ** p<0.05, * p<0.1		

Appendix E. Qualitative Study Details

Qualitative sample & data collection

To better understand the mechanisms behind our quantitative results we gathered additional qualitative data. The setting remains the same; the focus of this study is on the Prison Entrepreneurship Program in Texas.

We theoretically sampled (Eisenhardt, 2023) to ensure that we can explore the possible mechanisms of entrepreneurship versus employment and access to post-release resources (PEP only operates resources within Texas). Our goal was to understand *which* pieces of the program were useful in helping participants in not returning prison and *why*, not *if* the program was helpful. Thus, we focused our interviews on the population who had successfully avoided recidivating. Additionally, we attempted to balance the sample in two ways that theoretically may have been meaningful. First, we made sure to sample both participants who had and had not started a business since being released. This helps to ensure that we can account for whether the entrepreneurs and non-entrepreneurs are getting different things out of the training, despite a relative minority reporting having started a business. Second, we sampled both those who remained geographically close to the program (in Texas) and those who had moved to different parts of the country. PEP provides various resources after release, but the majority of those are only accessible in the vicinity of the program staff and network. Access to these resources may change how the participants thought about the program helping them.

Our main data source is interviews with former participants and program leaders. We were able to reach former participants directly through the PEP program as they maintain continued correspondence with many of their former participants. We interviewed 11 participants with interviews lasting between thirty and sixty minutes. Six interviewees had founded businesses, three were employed, and two were working in non-profits. We stopped interviewing at this point, despite having more interviews lined up, because we were consistently hearing the same types of answers over and over, signaling that we had reached theoretical saturation (Locke, 2011). We additionally conducted four interviews with program staff and leadership in order to get their perspective on how the program was run. A list of informants is

provided in Table E1. We complement the interview data with archival data from the program.

Additionally, we accessed testimonials from the program website which could be used to triangulate what our primary informants were telling us (Jick, 1979).

We used established procedures to ensure data validity while conducting these interviews, including employing interview techniques such as non-directive questioning that have been shown to yield accurate information from informants (Huber and Power 1985). The question began in a broad manner, allowing informants to tell their stories about participating in PEP and how it has helped them. Only later did we key in on particular aspects of the program with follow-up questions. This interview format allows informants to bring in any and all information they believe is pertinent, while also ensuring that all informants discuss similar topics eventually (Eisenhardt, 2001). The interview protocol is provided in Table E2. All interviews were recorded and later transcribed for accuracy.

Data analysis

Our analysis occurred in a series of iterations across levels of abstraction. Initially, each author coded about half the interviews independently. At this point we used “in-vivo” codes where possible, using informants’ own words and descriptions as part of the coding scheme (Locke, 2001), but kept topics in mind from theory both on prison training and entrepreneurial training. All three authors met at this point to discuss similarities and differences amongst our codes. We noticed that while we had sometimes focused on different phrases for the codes, we were encountering very similar themes in what we had coded for why the PEP training helped and how it changed the participants. Consistent with an active categorization process (Grodal et al., 2021) we began clarifying these second order themes and exploring how they may aggregate in parallel. At this point we could not come to parsimonious consensus, so we returned to the data individually, but with our evolving coding scheme in mind, and coded the rest of the data. Alongside this evolving coding scheme, we each individually memoed on our interpretations of stories from the interviews (Barley, 1990).

We iterated in this way multiple times, honing our first order codes, discussing second order themes, and categorizing and recategorizing themes into aggregate dimensions. For instance, at one point

in the process we had broken themes out between effects on character and effects on entrepreneurship, but the ties between themes were too strong in the data for this artificial separation, so we recombined them into three aggregate dimensions. We continued to refine our coding scheme until all three authors were in agreement, and then recoded our data again. See Table E3 for our final data structure and Table E4 for examples of our themes.

Table E1. List of Informants

Informant	Career history	Resides in Texas	Started Business?
NE1	Field supervisor of a nationwide company doing flooring	Yes	No (employed)
NE2	Started LLC to sell t-shirts	Yes	Yes
NE3	Worked in John Deere dealership, became Level 3 technician. Moved to Missouri for salaried construction work. Trying to start his own company as construction handyman, got business cards, but still working on salary.	No	No (employed)
NE4	Went home to Amarillo TX after 6 months in Dallas halfway house. Fell back on old habits and started using drugs. Moved to Montana in 2016. Clean for 5 years now, so since 2018. Works in HVAC business	No	No (employed)
NP1	Became a servant leader and a PEP employee for 5 years. Promoted to executive relations manager in Houston. Joined a grant writing firm with PEP contact/boss, worked with nonprofits. Got his bachelor's degree in U of Houston. Worked for Habitat for Humanity, American Heart Association. Now working for a large oil and gas company leading social impact. Thinking about going into nonprofit consulting.	Yes	No (nonprofit employee)
NP2	Worked with young adults who have been impacted by gun violence. Lead for critical response. Worked with other orgs to create resources and mentorship. Now teaching nonprofits how to work with young adults.	No	No (nonprofit employee)
E1	Started 5 companies	Yes	Yes
E2	Took certificate courses, Got an UG business degree after, focused on sales. Got a partner and opened a barber shop. Now does marketing consulting for small businesses in the area. Doing a double major in fashion marketing	Yes	Yes
E3	Used to be a drug dealer. Worked in trucking, as fleet mechanic, construction. Owned contracting company, got into real estate	Yes	Yes
E4	Had to go to a different city. Got a degree in ministry. Starting own nonprofit with spouse on marriages impacted by incarceration	Yes	Yes
E5	Used to sell drugs. Upon release went to school to become a realtor. Represented himself in court for his license. Got his real estate license, started his own company	Yes	Yes
PE1	Works in program. Was servant leader, Became peer educator/manager.	Yes	No (program employee)
PE2	Went to Austin, worked at PetSmart, became house manager within 2 weeks, later was hired by PEP as a case manager. Started business in fitness about two years ago, still continuing to thrive.	Yes	Yes (program employee)
PE3	Volunteered with Texas Department of Corrections in 2003 and then PEP in 2006. Not graduate.	Yes	No (program employee)

Table E2. Interview Protocol

- Why did you choose to participate in PEP?
 - *Was starting a business part of the reason you applied to PEP?*
- Were their skills or resources you were hoping to get through the program?
 - *What resources did you hope to access? (money, network, certificate, mentors...)*
- Was the program challenging for you? If yes, in what way?
- What was the most surprising part of the program for you?
- Do you believe PEP changed you in any way? How?
- Tell me about your employment or entrepreneurial experiences after you participated in the program, from the time you finished until now.
 - *[If no business] Have you considered starting a business? What stopped you from doing so?*
- *What challenges did you experience upon release, especially with regards to employment or entrepreneurship?*
- Do you believe PEP changed your post-release experience in any way? How?
- What pieces of the program have you used since your release, if any?
 - *Program components in the annual survey: Effective Leadership, Men's Life, PEP 10 Driving Values, ToastMasters, Entrepreneurship Teaching, Business Plan Writing, Personal Finance, Personality Indicator/MBTI*
 - Can you give specific examples of how you used [X]?
- Do you stay connected with the program? If so, in what ways?
 - What would you advise to someone considering joining the PEP program?
 - What actions may improve program outcomes for program participants?
- We have found that participation in PEP reduces recidivism, do you have any opinion on why that might be?
- Is there anything else you'd like to tell me about how PEP has changed you or your experiences?

Table E3: Data structure

First order codes	Second order themes	Aggregate dimensions
Desire to make an honest living, live a moral life, businessmen have integrity, etc.	Living with integrity	Entrepreneurship as vision
Businessmen as role models, seeing examples of different lives, observing the ideas/goals of others, etc.	Exposure to possibilities	
Who they want to be as a person (identity), what they can envision for their lives, goals for success, etc.	Future self	
Being on the lookout for new possibilities, seeing a second chance, etc.	Imagining possibilities	
Being more analytical, playing devil's advocate, being curious, etc.	Evaluating opportunities / Questioning	Entrepreneurship as focused planning
Reflecting on own personality, purposely slow down to observe world or their own actions, knowing when to ask for help, etc.	Self-reflection / Observation	
Knowing which skills they have, recognizing particular uses for skills like Toastmasters, financial literacy, etc.	Practical skills	
Desire to learn new things	Learning orientation	
Building a plan to get toward goals, determining a good way to fulfill a desired identity, step-by-step hypothetical reasoning, etc.	Route planning	
Taking responsibility for one's own actions (past and present), feeling like they need to do well to not let down brothers or community, etc.	Accountability	Entrepreneurship as control
Being good enough, thinking they will succeed, belief they can do difficult tasks, etc.	Confidence	
Acknowledging they control outcomes (good and bad), desire to be a boss/owner, self-sufficiency (financial / life), etc.	Ownership	
Nothing will stop me, overcoming challenges, pushing back against stigma, going outside your comfort zone, etc.	Resilience	

Table E4: Examples of aggregate dimensions and second order themes

Aggregate dimension	Second order theme	Exemplary quote(s)
Entrepreneurship as vision	Living with integrity	"I just learned to be a better person, you know that it's." "It really is breaking down your mindset, understanding. Hey, you know what that yes, we are teaching new business, but there's also the aspect of hey, you need to understand your character, your character development, understanding that to be a business owner and be a successful business owner." "Yes, you can learn business. Yes, you can learn all, you know, how to do all types of things, but really in essence, if you're not changing your character, you're really just becoming a smarter criminal..." "It's not just about being an entrepreneur, it's about how to live a good life, a better life." "To be a business owner and be a successful business owner, you have to have these qualities of accountability and integrity and excellence in pursuing all that." "The whole thing behind that, it's like OK, if you cannot manage your own life, if you can't keep your life on track, if you can't be on a straight path, how can you, one be a good employee for anybody else, or two even own and operate and manage a small business and those people under you?" "I had all the intentions in the world of going right back and doing what I was doing, but doing it bigger. And better. And I even told my family that... They were like, do you regret going to prison? And I was like, this is just an occupational hazard. It's just going to happen. Like I was dead set on doing it the easy way. But then there was a point... I remember we were all in line to go to chow. And someone told some joke and it would have been hilarious any other time. And then even the guy who told the joke was like, yeah, I'm not proud of that joke. And he's like, it's weird that would have been funny six months ago..."
	Exposure to possibilities	"It changed the direction of my life because it exposed me to something different. It exposed me to want to try to reach for something that I believed before that I was unable to grasp."
	Future self	"It's challenges us as we go through the program to dig deep inside and to think about life in general, not just business. Hey, you want to be successful? But what does your life look like? What do you want your life to look like aside from this? ... Once you visualize that and you see yourself succeeding then it becomes more realistic for you." "The guy that walked in was just sharp, walked in, in a nice suit with a briefcase, and I'm thinking, OK, this is, you know, very well, educated well put together guy... And then he showed us a video and in the video that he showed us, we saw that same guy in a white jumpsuit. And I was blown away. I'm like, like, right in front of me, I saw the transformation and I saw the opportunity for myself. Like, man, this guy could do it. Surely, I can do it too." "My business plan was I wasn't very creative and just I was aimed kind of low and it was towards like a lawn care business. And I

		just, I wasn't going to do that. But there was other people who had a lot of good business plans. And I was like, that's a great idea and one was like a truck driving school. And I got he was he had a CDL and he told me.”
	Imagining possibilities	“Showing somebody that there is a light. And there is second chances.” “The fact that I'm here, that I have this opportunity, yes, that's a reminder every single day of the things that I learned...To become a real estate agent is crazy. I've never in my whole entire life would ever, ever think that [was possible]. It would never cross my mind, ever, ever, ever. And I'm a good real estate agent.”
Entrepreneurship as focused development	Evaluating opportunities / Questioning	“But then I found out pretty quick by talking all the truck drivers that that wasn't really the best business to be in anymore. And uh, I met somebody, my other, a friend of mine got out and he was working. I wanted to hang out with him... but he was working on building a pergola - like a thing to cover your backyard - and I always had tools and I was working always doing things. But like we, we built it. The guy who hired us, screwed the whole thing up and I had to fix everything. And I was able to and then I found out that he was making \$7000 on that thing and it was like \$7000? And this guy doesn't know how to do it? And I was like, oh, and I realized like, oh, because they teach you to recognize opportunities. So I was like, oh, there it is.” “I think I used a lot of it throughout my life. Since then, you know, I mean in the way that I go about dealing with things and approaching things. I'm more analytic about things. Instead of just jumping in and doing something, I consider the cost and what the repercussions are and the outcome of the different situations that I find myself in.” “Yes sir. I mean, I would have probably got started... Before I do it I want to make sure that I'm ready for it.”
	Self-reflection / Observation	“One thing that I remember a lot of like specifically, I remember them talking about is like seeing what do we need to work on. Where are we seeing the flaws? Where in our own life can I take an honest look?” “I need to do something, invest in things that is going to help me further in advance in my life and my career. In my finances, with my family, to provide a stable foundation for us because I wasn't used to having the cushion to fall back on if the economy went down or I lose my job. I didn't have that before. PEP taught me that, you know, when we go through the whole finance stage of it.” “Remember, your life is about 5 miles an hour. Because at 5 miles an hour you are able to see everything and anything around you you're able to be aware of what's going on. At 100 miles an hour, you're moving too fast and you can't see that speed bump that's coming to you. And by the time you see it, you gotta hard brake. Now you probably gonna mess up... at 5 miles an hour it's a smooth jump over that.”

	Practical skills	“They gave us and like of, you know, interviews and how to write a resume and just the budgeting. And like some really good life skills.” “So it sticks out to actually be able to write and be sensible in a business sense. It really stands out to people when you're in a job and you're able to look professional and do the things that PEP taught. Like I can use most all of it... the business plan writing I use to do like say when I want to present an idea to someone, then I'll use that form and I'll write it up or print it out.” “We have everything from naming the business to marketing, You got to learn it all. It gave me enough smarts to have a conversation with somebody and not sound like I don't know what I'm talking about. So I can sit down with realtors, at and sound like a businessperson.”
	Learning orientation	“As I was working full time [detailing]... I was still interested in learning the sales portion of things. We do through PEP, we did go through a segment in there of, you know, integrity selling. But to actually now understand, you know what those terms mean and to actually realize it in this situation. I was very interested. So I started learning more about sales, learning how they in particular, Ford, I was at a Ford dealership, how they want you to approach sales, inventory and understanding your product. And they offer courses on there and so I made an attempt on my end.”
	Route planning	“When you have, when you have a goal and you have a plan and you're sticking to it, it was really kind of a breeze. It made it look easy. Feel easy.” “Knowing the processes of everything and doing your research before you dive into something because there's a bunch of business owners and people that start small businesses that just dive in because something intrigues them. So yeah, I want to be a part of that too. That without doing research and you know, within five years they fail because the preparation wasn't there. So when I went through the program was back in 2006. So from 2006 up until now has been preparation and waiting for the opportunity to present itself and not just diving into something. So you know, doing the proper research and preparation before actually stepping forward [is] a huge thing [t]hat I learned also.” “Well, I've always been able to get a good amount of money, like I've never like, you know what I mean? I've always had plenty of money, I've always been able to pull that off. But then it would all like I wasn't going to do nothing with it. I would just and then easy, easy come, easy go. And then you'd have like, 20 grand. And I couldn't even find an apartment to rent because he can't just show up with money and no, no job experience and. But they taught me the importance of my credit getting right. Once you get the, then I just got the credit. Right. Got this right. Before, I never had a plan. You have to have a plan, like some kind of plan. You got to be working towards some kind of a goal overtime and then even if it even if the plans change, you're still on track for something.”

Entrepreneurship as control	Accountability	“It pushed me because I realize, you know, instead of always laying things on other people, you know, why don't I go out and try my own? And you know, at the end of it, if I fail then, I know that I failed. But I'm not blaming it on someone else. If I failed it's all me and that's kind of what I like about. that idea of owning my own business. I mean, mostly because I'm a human. I never wanted to fully take credit for mistakes, so I always blamed others.” “It's not just about skills, or keeping people out of jail, it's about accountability.” “You have to have your shirt tucked in. You cannot cuss. You have stuff to do from morning till night and you have to be accountable for any and everything you do wrong. And they don't kick you out. They just make you quit. So if you do something wrong, it's not like they kick you out. If you do something wrong, they'll just. Give you like a dictionary and you like copy that by next week... the whole dictionary or whatever. And if you can't do it then then you're then you're not in the program anymore. But if you really want to be in that program, you take responsibility.”
	Confidence	“They gave me the confidence to maybe want to start my own business and be successful just in, like, my job.” “Overall you know giving the confidence to go out there, rock an interview and rock your resume and get it.” “OK, I'm gonna be the best whether it's washing dishes or whether it's running being a CEO. Like PEP kind of gave you that confidence and I'm gonna go rock it.” “And like I got out on fire, like I was so full of all the info that they had given me, all the confidence that I had gained by giving speeches to people that loved it, you know, like not getting this the confidence that that I built up.” “I appealed the decision then I cause I wanted to explain myself. You know, I want to give myself a fighting chance. So when I went to appeals, the Texas Real Estate Commission, they brought their lawyers, two lawyers. They brought their two lawyers and I went by myself and this... when you go to PEP, you got to go in what's called Toastmasters, which, you know, shows you how to address a room properly. You know, how to address the room and how to speak properly.”
	Ownership	“You know, I've been given a I've been given helping hand up and not hand out. And I'm very, you know, appreciative of that.” “For me, now at this point because of PEP and what I got out of that, my life is just way too good to be even considering like anything criminal. Until you, until you're financially free from the... I don't want to call it the system... But once, once you get out of that rat race, that's what true freedom is like. Like I can... I'm at work right now, but I can just stop right now and if I want to.” “It's one of those things to where we could provide you with the tools, but it's up to you to pick up the tool and use it and that's what they do.”

		“Oh yeah. So, entrepreneurship is being your own boss, taking ownership of who you are, and leading your own life. It's not just about business. Entrepreneurship is being the decision maker, making your own decisions. That's something that really stuck out. Entrepreneurship wasn't about just running a business. It's about learning how to run your own lives.” “One of the biggest things that we emphasize, is ownership; taking ownership of you, your life, and not allowing circumstances and other people to dictate your life. You have to take ownership of your life. You've got to take responsibility. You have to be the decision-maker. You know, life is 10% what happens to you, 90% of how you respond to it, and so many people are moved by what happens to them. Entrepreneurship has to deal with that. You've got to be the decision-maker. You've got to be the one to run and own your own life.”
	Resilience	“Like I'm stronger... all that time of being out of my comfort zone and it just strengthened me... I'm willing to experience more than I ever would have. You know I'm very open and I have less fear of things. You know, when it comes to experiencing life... It strengthened my mentality of going out and exploring things. If it's just fear that's stopping me from something, I'm gonna jump into it. Just because it was like the dancing thing, I was scared to death of dancing in front of all the people. [Now]I dance regularly. I love it. I love to dance and I love to sing. And I, I mean, I know it's a goofy thing.” “It truly changed my because I was the type of person before where if I saw a task too hard I would give up. I'm like whatever. I'm like, man, that's too hard to even try I'm not gonna try that no more. Forget that. It's too much work to go into it. And PEP helped me grow in that aspect.”

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